WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER PROGRAM

AZNS-127 (PAD #54), TYPE: IIF

Drilling Department

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MSA/OEOC			
	Position	**Name**	**Date & Signature**
Prepared by	Drilling Engineering Department	Masoud Babakhani M.H.Haji Taleb	
Approved by	Project Head of Drilling Engineering	Mostafa Khedri	
	Project Head of Drilling Operation	Piran Farzaneh	
	Project Manager	Rashid Ansari	
PPL			
	Position	**Name**	**Date & Signature**
Check & Approved by	Drilling Operation Manager	Majid Nariman	
	Subsurface Engineering Manager	Mahdi Salimmehr	
	Project Manager	Ali Maghzi	
	Subsurface Project Manager	Alireza Hamoudi	

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

This Document covers the work over of well AZNS-122. Main headlines of program are de-complete the well and run new completion string.



Figure 1: AZNS Oilfield Location Map

1.2 Well Information

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type Of Operation:	Workover
Well Name / Type :	AZNS-127 / Fahliyan well in Block II
Pad Number:	54
Well Type:	Vertical
Wellhead Coordinate:	N: 3,448,561 E: 196,167
TD Coordinate:	N: 3,448,538.7 E: 196,163.4
Well Objective:	De-complete the well, Re-complete the well, Due to abnormal annulus pressure in 9-5/8" CSG X 4-1/2" TBN annulus, find leakage point & cure it, Check and smooth collapse interval from 3889-3911
RTE:	13.45 m
GLE:	3.95 m
Well Duration :	60.4 (Main Plan) + 58.9 (Contingency Plan)
Total Depth (MD):	4226 m
Producer Formation:	Perforated in 7" liner (Exiting Perforation TCP : 3956-3977 m, 3995-3998 m, 4043-4049 m. WCP: 4071-4077 m, 4026-4023 m, 4005-3999 m, 3999-3998 m, 3995-3993 m, 3993-3987 m, 3987-3984 m)
Formation TD :	Fahliyan
Reservoir :	Fahliyan

Table 1: AZNS-127 information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.
Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.
Block II is located between well AZN-010 and AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal
Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure and all drilled wells in this block suffered from complete loss at Pabdeh formation.

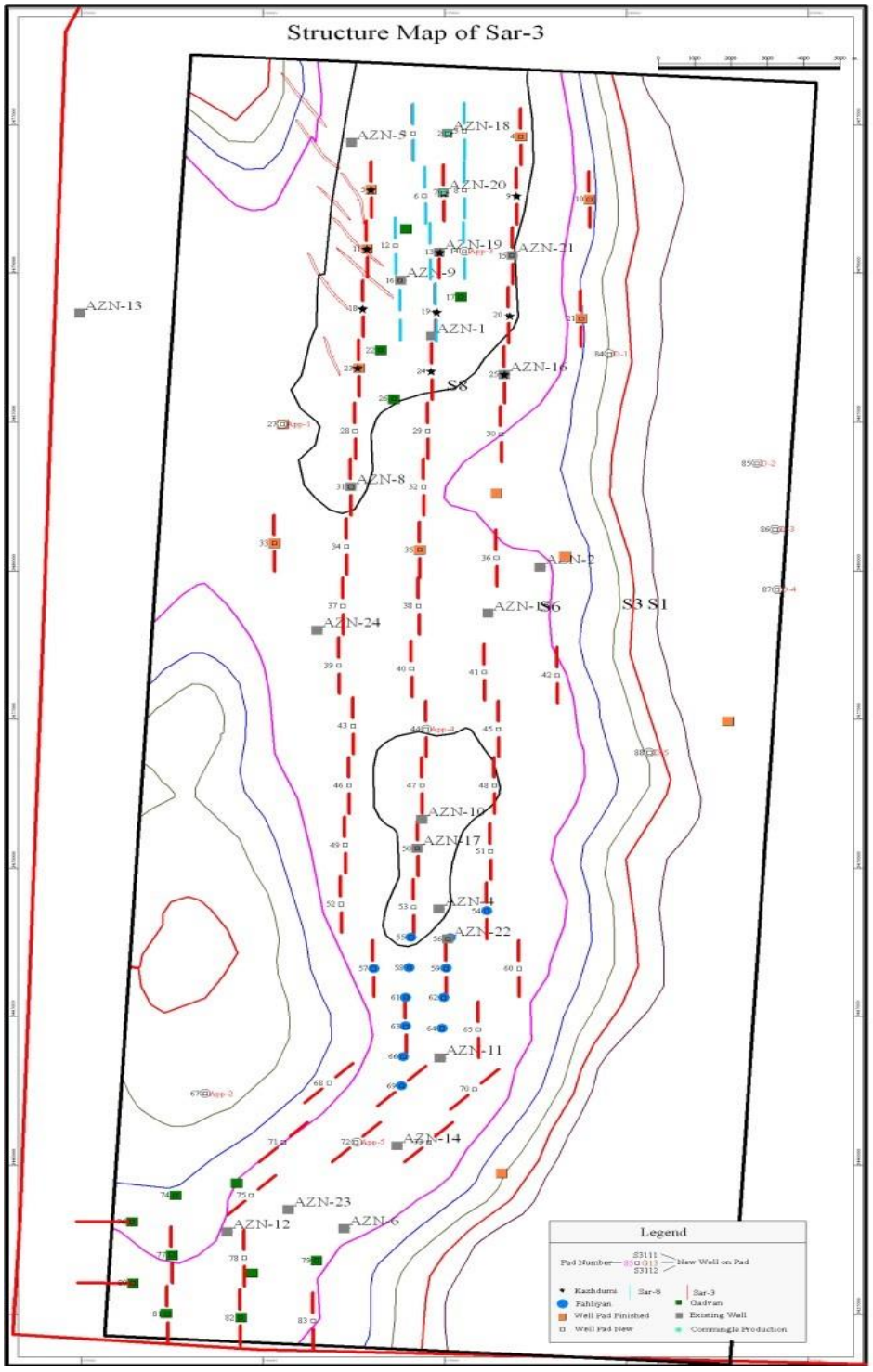


Figure 2: AZNS Area Distribution

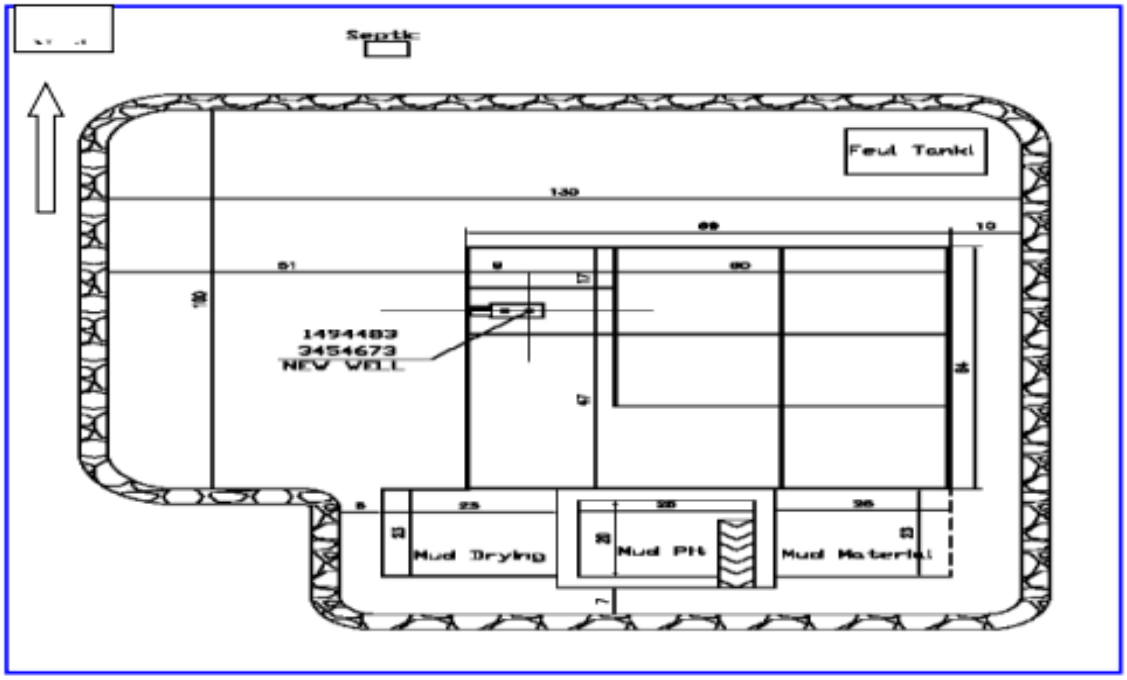
1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

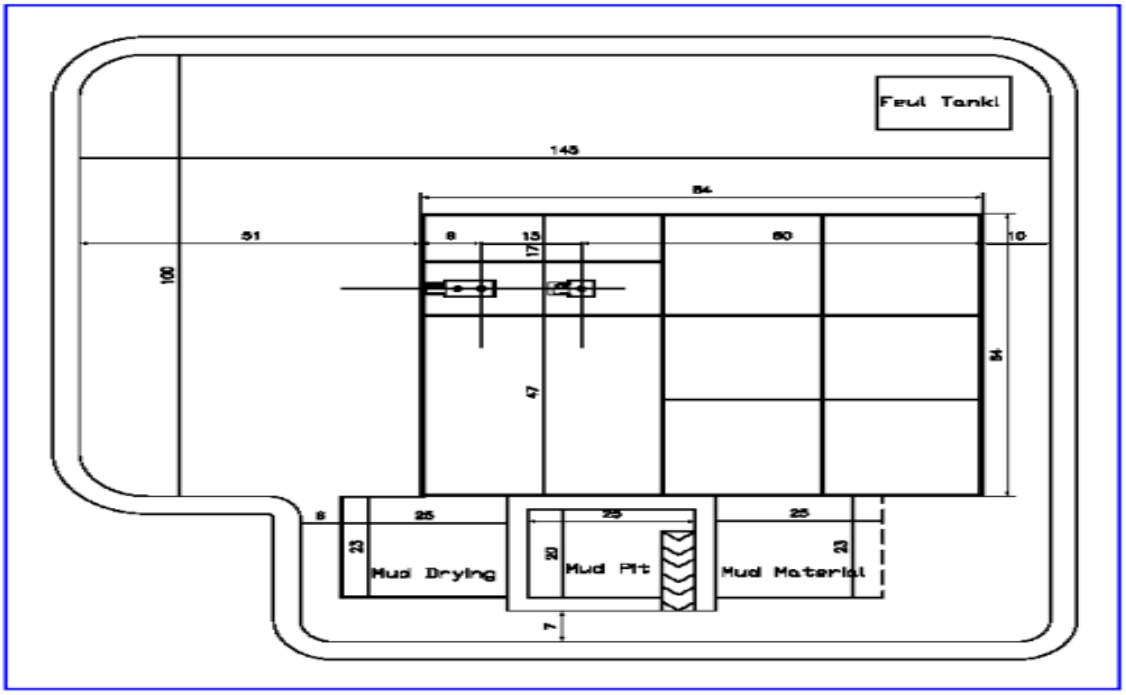
Table 3: Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 kg/cm2
• Fuel tank concrete pond dimension: Length 20m×Width 12m×Height 0.5m
• Drainage: Width 30cm × Depth 30cm
• Burning pit: 15m×15m square, located on South-East or South-West with at least 350m distance from well head
• Cellar to cellar distance in multi-well well pad: 15m
• Conductor pipe depth under cellar bottom is 5.5m. The conductor pipe will be driven while constructing the well pad.
• Mud material dimension is 25m*35m +granular surface
• Mud drying dimension is 25*35m + concrete surface
• Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

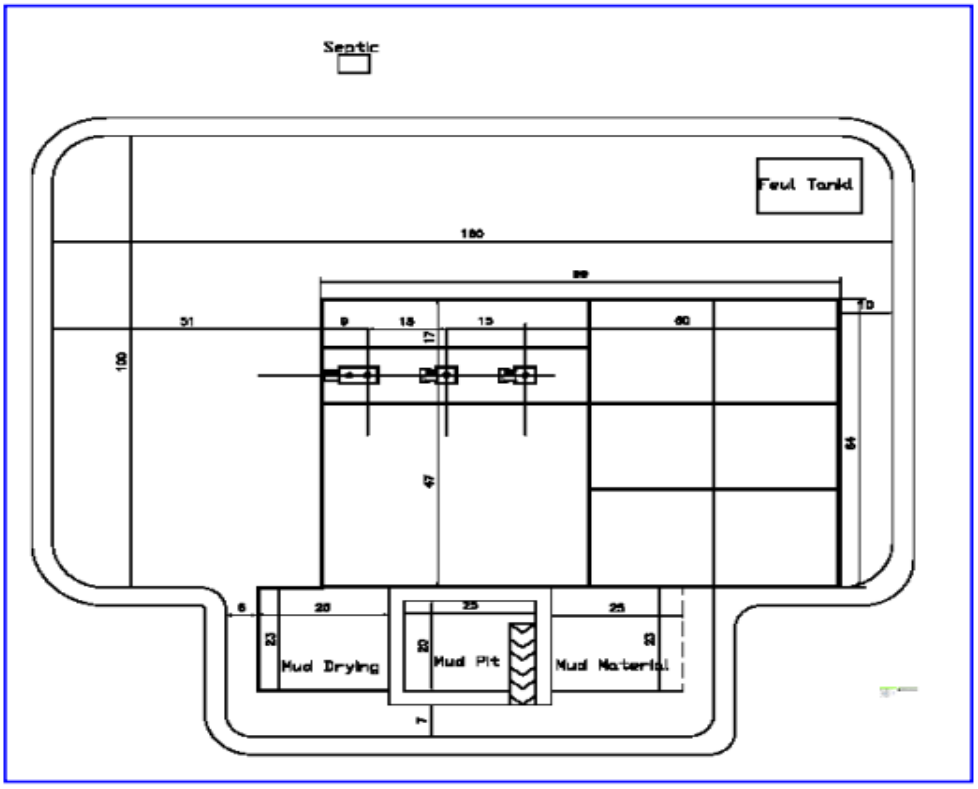
1. Types of well pads:
• One well in one pad.
• Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
• Three wells in one pad: 2 Sarvak horizontal wells and 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads Sketch

1.5 Reservoir Data

Reservoir Name:	Fahliyan	Unit
Fluids:	Oil	….
Datum Depth:	-4000	M,TVDss
Datum Pressure:	Fah-4a= max 6800 psi Fah-5a= max 7400 psi	Psi
Pressure Gradient:	0.28	Psi/ft
Temperature:	132	C°
Temperature Gradient:	N/A	1.32 F/100ft 0.73 C/100ft 2.4 C/100m
H2S:	H2S in Gas at Atm condition 3000 PPM	PPM
CO2:	Co2 in Gas at Atm Condition 50000 PPM	PPM
Salt Salinity :	143000	ppm
Calcium Content:	8750	(mg/lit)

Table 4: Reservoir Data

Reservoir	Sub-layer	Mid of Sub-LayerDepth in AZNS-122 (mss)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure
MDT 195 MDT 195 MDT 195	Fah-1a	-3966.7	256.3	1.32	9100	MDT 195
	Fah-1b	-3983	257		9100	MDT 195
	Fah-1c	-4009	258		9200	MDT 195
	Fah-2a	-4035.7	259.3		8912	Dynamic Model
	Fah-2b	-4049.8	259.9		8924	Pressure Gradient
	Fah-3a	-4058	260.3		8440	Dynamic Model
	Fah-3b	-4067	260.6		8447	Pressure Gradient
	Fah-4a	-4077	261.1		7551	Dynamic Model
	Fah-4b	-4091	261.7		8100	MDT 195
	Fah-5a	-4108.6	262.4		8100	MDT 195
	Fah-5b	-4163.7	264.8		8100	MDT 195

Table 5: Predicted Pressure and Temperature and TTZ Contact

1.6 Formation Pressure Prediction in Block II

Predicted Formation Pressure in Block II				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0-1095.9	0.98-1.18	0.06-1.39	1.01-1.89
Gachsaran	1095.9-1461	1.13-1.31	0.78-1.36	1.89-2.01
Asmari	1461-1728	1.14-1.26	0.6-1.28	1.43-1.79
Pabdeh	1728-2196	1.20-1.26	0.82-1.36	1.40-1.78
Gurpi	2196-2536	1.21-1.26	0.66-1.02	1.40-1.76
Ilam	2536-2626	1.18-1.24	0.20-0.9	1.41-1.72
Laffan	2626-2633	1.13-1.18	0.24-1.11	1.41-1.72
Sarvak	2633-3278	1.14-1.26	0.24-1.23	1.44-1.81
Kazhdumi	3278-3481	1.1-1.14	0.46-0.96	1.93-1.94
Dariyan	3481-3647	1.18-1.25	0.55-1.1	1.96-2.06
Gadvan	3647-3937	1.1-1.25	0.55-1.44	1.97-2.11
Fahliyan	3937-T.D.	1.60-1.3	1.20-1.75	2.08-2.12

Table 6: Formation Pressure Prediction

1.7 Drilling History

26" Hole Section

Rig moving initiated on 95.09.27 and spudded rig 304 on 95.10.08 at 1200hrs.

Commenced drilling 26" hole section from 14m to 150m while experienced low ROP and bit balled up several times. Pulled out to surface, Ran in hole with same rock bit and 26" reamer to 150m and reamed from surface to 150m. Ran 20" casing as following with no major problem:

- 20" stab in shoe 94ppf, k-55, BTC (KTK Co.)
- 2 joints 20" casing 94ppf, k-55, BTC to 25m
- Circulated mud and check float shoe-ok
- Continued run with 10 joints 20" casing 94ppf, K-55, BTC to 146m (Totally run: 12 joints 20" casing 94ppf, K-55, BTC) to 146m. Cemented 20" casing in place as below:
- Sting in, lined up cement line and tested it with 2000psi-ok.
- Pumped 20bbl fresh water as spacer, mixed and pumped 689sxs cement class "G" Kerman (slurry weight: 118pcf, slurry volume: 170bbl), displaced with 7bbl 68pcf mud (1.7bbl short displace).

Note: Got 40bbl cement slurry and 10bbl contaminate as return on surface.

17 1/2" Hole section

Drilling of 17-1/2" hole section started on 12th Dey 1395 by new 17-1/2" rock bit and 17-1/2" string stabilizer placed at 20m, from 150m to 1064m while pumped salt saturated water several times to prevent balled up bit and displaced hole with 74pcf Kcl polymer mud at 502m.

Continued drilling 17-1/2" hole section by 17-1/2" from 1064m to 13-3/8" casing point at 1441m with 3-6bph loss while pumped hi-vis pill several times and pumped salt saturated water several times to prevent balled up bit. Condition hole prior logging, pulled out to surface and ran in hole with Sonic/Caliper/CDR/GR wire line logging tool string to 1438m (3m off bottom), took weight, tried to pass this point-not succeed, logged up from 1438 to 1280m, due to failure of Sonic tools, stopped log up and pulled out of hole to surface, changed sonic tools and ran in hole with same logging tool string

to 1438m (3m off bottom), tried to pass this point-not succeed, logged up from 1438 to 146m, continue logged up with GR to surface.

Rigged up 13-3/8" casing running equipment, held PJSM and ran 13-3/8" casing as following:

- 13-3/8" float shoe (KTK, 68ppf, N-80, BTC)
- 2 joints 13-3/8" casing 68ppf, N-80, BTC as shoe track
- 13-3/8" float collar (KTK, 68ppf, N-80, BTC)
- 2 joints 13-3/8" casing 68ppf, N-80, BTC, circulated mud & tested float shoe & float collar- ok.
- Continued RIH with 13-3/8" casing 68ppf, N-80, BTC (Totally:127 jts) to bottom at 1441m while filled up string every 10 joints& installed centralizer as program, tagged bottom, pulled out to 1440.5m, circulated mud as 1.5 cap of casing with 1-2 bph loss.

Cemented 13-3/8” casing in place as below:

- Pumped 40bbl chemical wash.
- Pumped 80bbl 93pcf spacer.
- Mixed & pumped 2004sxs Cement “G” Kerman as lead (slurry volume: 780bbl, slurry weight: 98pcf).
- Mixed & pumped 582sxs Cement “G” Kerman as tail (slurry volume: 142bbl, slurry weight: 118pcf).
- Dropped top plug (KTK).
- Displaced with 696 bbl 90pcf mud, bumped plug with 1500psi and held pressure for 15 min-ok. Note: Got 150bbl cement slurry and 120bbl contaminate as return on surface and recorded 40bbl loss during cement job.

Ran in hole with 12-1/4” used R. bit and drilled out top plug, float collar, medium to hard cement of shoe track and float shoe to 1441m while worked on junk sub several times.

Note: Confirmed float collar at 1417m and float shoe at 1441m

12 1/4" Hole section

Commenced drilling 12-1/4” hole section from 1441m to 1447m, displaced hole with 80 pcf clay-bond drilling fluid.

Pulled out to surface to change bit and BHA (recovered 4lb metal as junk), ran in hole 12- 1/4” PDC bit and 8” PDM (AKO:0 degree) and continued drilling 12-1/4” hole as performance drilling from 1447m to 1612m while periodically pumped tandem pill with 3-5bph loss. Due to APEK PDM failure, pulled out of hole to surface to change PDM, ran in hole to bottom and resumed performance drilling 12-1/4” hole from 1612m to 1989m with 3-5bph loss while from 1750m periodically pumped FL controlled brine and also pumped hi-vis caustic pill several times.

NOTE: From 1750 m due to change lithology to Arg. limestone, ROP decreased, tried to increase ROP by change in drilling parameters with no success.

Due to pressure dropped from 2100 to 1950psi, stopped drilling, checked surface equipment and pulled out of hole to check string, on surface observed wash out on XOS (6-5/8" Reg. pin x 4-1/2" IF box) between 5" HWDP and 8-1/2" DC.

NOTE: Cherty zone from 1920 m.

Continued drilling 12-1/4” hole from 1989m to 2000m while periodically pumped hi-vis. Pulled out of hole to surface, P/U and M/U 12-1/4” PDC bit to 8” PDM and ran in hole to bottom, resumed performance drilling 12-1/4” hole from 2000m to 2354m, suddenly pressure dropped from 2850psi to 1950psi (900psi), stopped drilling, checked surface equipment, pulled out of hole to surface to check string, on surface observed float sub twist off as (left in hole : 12-1/4" pdc bit, 8" PDM, 0.36m form float sub, length of fish : 9.18m, top of the fish :2344.6m).

P/U & M/U 10-1/8" over shot dressed with 8-5/8" control, 8" spiral grapple and RIH to top of the fish at 2344.6m, engaged and catch fish, pulled out to surface while completely recovered fish.

Ran in hole with 8” PDM to bottom and resumed performance drilling 12-1/4” hole from 2354m to 2371m with 2-3bph loss, due to low ROP stopped performance drilling and pulled out of hole to surface and L/D 8” PDM.

Ran in hole with 12-1/4" B. bit and continued drilling 12-1/4" hole section from 2371m to 2383m, due to low ROP pulled out of hole.

P/U and M/U 12-1/4" PDC bit and 8" PDM and RIH to bottom and resumed performance drilling 12-1/4" hole from 2383m to 3090m while periodically pumped hi-vis pill with 2-6bph loss.

NOTE:

- From 3005m to 3021m, due to change lithology to limestone-shale, decreased in ROP and increase loss to 10bph.

- From 3030m to 3090m due to increase in loss to 7-9bph, spot 70bbl 5 ppb Shell fish LCM pill.

Continued performance drilling 12-1/4" hole from 3090m to 3117m with 9-10bph loss, due to pressure dropped from 2560psi to 2320psi, stopped drilling, checked surface equipment and pulled out of hole to check string, string got at 3040m, worked on stuck with downward & upward by applying maximum 50klbs over pull & pumped 40bbl hi-vis pill several times, pipes got free and continued pulled out of hole to surface. On surface observed wash out on float sub above 8" PDM.

Ran in hole with 12-1/4" PDC bit to bottom and continued drilling 12-1/4" hole from 3117m to 3327m with 6-9bph loss while several times pump 7ppb hi-vis LCM pill (4ppb shell fish+3ppblimestone) and also 16ppb LCM pill (5ppb shell fish med.+7ppb limestone med.+4ppb limestone coarse). At 3327m loss increased from 6-8bph to 15bph, to pump LCM pill pulled out of hole from 3327m to 3050m, spot 100bbl hi-vis LCM pill 20ppb (10ppb shell fish med+10 ppb limestone coarse), ran back from 3050m to bottom at 3327m and hesitated for 1 hr with 1bph loss.

Continued drilling 12-1/4" hole from 3327m to 3705m with 5-9 bph loss while pumped hi-vis 20ppb LCM pill (10ppb shell fish med+ 10 ppb limestone coarse) several times. At 3705m loss increased to 12bph, decreased flow rate from 650gpm to 550gpm & pumped 20ppb hi-vis bentonite LCM pill (10ppb shell fish med+10 ppb limestone coarse).

Continued drilling 12-1/4" hole size from 3786m to planned 9-5/8" casing shoe at 3910m with 5-9 bph loss while periodically pumped 15ppb hi-vis bentonite LCM pill (7.5ppb shell fish med+7.5ppb limestone coarse).

Rigged up logging equipment and ran in hole with first full suite tool string as: MSFL/DLL/SONIC/CALIPER/GR/CDR tools to 3900m, string held up, tried to pass this point with no success, logged up from 3900m to 13-3/8" casing shoe at 1441m, continued POOH to surface and laid down logging tools while decided to repeat DLL log.

Ran second full suite logging tool string as: ZDL/CNC/DSL/WTS/TTRM to 3855m (confirmed by PEDEC witness), logged up from 3855m to 13-3/8" casing shoe at 1441m, continued POOH to surface and laid down logging tools w/1-2 bph loss.

R/U third logging tool string as: DLL/GR, ran in hole to 3630m (confirmed by PEDEC witness), again recorded resistivity log data as client request from 3630m to 1441m and logged up logged up from 3630m to 2150m, and continued pooh to surface and R/D logging equipment.

Ran in hole with 12-1/4" B. bit to bottom at 3910 and pulled out of hole to 13-3/8" casing shoe at 1441m while W & R 3890m to 3910m. Circulated mud at 13-3/8" casing shoe while waited on receive 9-5/8" casing. After 69hrs NPT, received 9-5/8" casing, to condition hole prior run casing RIH to bottom and pulled out of hole to surface with good hole condition.

Rigged up casing running equipment and ran 9-5/8" casing as following:

- PJSM
- 9-5/8" float shoe KTK, 53.5ppf, C-110, TSH Blue
- 2 joints 9-5/8" casing 53.5ppf, T-95, TSH Blue as shoe track.
- 9-5/8" float collar KTK, 53.5ppf, C-110, TSH Blue.
- 2 joints 9-5/8" CSG 53.5#, T-95, TSH Blue.
- Circulated mud & function test float shoe & float collar-ok.
- Continued RIH with 99 joints 9 5/8" casing 53.5ppf, T-95, TSH blue (totally 103 joints) to 1163.6m.
- 9-5/8" XOS TSH blue pin x TPCQ box,

• Continued run with 9-5/8" casing to bottom at 3910m with good hole condition while washed down from 3900m to 3910m while and circulated mud at 1441m & checked casing fill up every 10 jts- ok.

• Ran totally: 103 joints 9-5/8" CSG 53.5ppf, T-95, TSH Blue + 9-5/8" XOS TSH Blue pin x TPCQ box + 226 joints 9-5/8" CSG 53.5ppf, T-95, TPCQ. Circulated mud with 4-5bph loss at bottom while waited on receive cement & additives by NIDC for 65hrs, casing got stuck and unable to move upward and downward.

R/U cement equipment and pressure tested to 4500 psi-ok, cement 9-5/8" casing in place as following:

• Pumped 20bbl 85pcf spacer.

• Lead: mixed & pumped 2282sxs cement "G" delijan with cenosphere, micro silica (slurry volume: 1133bbl, slurry weight: 88pcf).

• Tail: mixed & pumped 363sxs cement "G" delijan with o-silica flour (slurry volume: 116bbl, slurry weight: 118pcf).

• Dropped top plug (KTK), displaced with 902bbl 81pcf mud, bumped plug with 3000psi and held pressure for 15 min-ok.

• After 702bbl displacement, observed cement slurry on surface (got 200bbl slurry cement & 22bbl contaminate on surface), recorded 280bbl loss during job.

• Unable to cut landing joint due to back flow**.**

NOTE:

• After 8hrs opened cement head valve & observed back flow, closed valve, again after 6hrs opened valve & check again, due to back flow closed valve & waited on cement to get hard.

• After 10hrs tail surface sample got hard completely.

• After 41hrs lead surface sample got hard 40%.

P/U and M/U 8-1/2" B. bit and ran in hole from 3700m to 3852m, tag top of cement at 3852 m, drilled out medium to hard cement from 3852m to 2m above 9-5/8" shoe at 3907m. Displaced hole

with 104 pcf ultra-carb drill fluid and continued drill out medium to hard cement from 3907m to 3910m while drilled out float shoe at 3909m and worked on junk sub several times.

Note:

- Confirmed 9-5/8" float collar at 3886m and 9-5/8" float shoe at 3909m.
- Recovered 1.5lb metal wool from junk sub

8 1/2" Hole section

Commenced drilling 8-1/2” hole section drilling from 3910m to 3913m, circulate mud and performed FIT as per program by applying 270psi pressure and hold it for 15 min-ok. Pulled out of hole to surface.

Ran in hole with 9-5/8” scraper and scraped from 3600m to 3909m, rigged up logging equipment, ran in hole with logging tool string as: VDL/SBT/CCL/GR, string held up at 3905m, tried to pass several times with no success, logged up from 3905m to 100m above of 13-3/8"casing shoe at 1341m, continued pull out of hole to surface and rigged down logging equipment.

Ran in hole with 8-1/2” PDC bit and commenced drilling 8-1/2” hole section from 3913m to hole TD 4226m with 0-3bph loss with no major problem. Pulled out of hole to surface while condition hole prior run log.

Note: Due to observed tight hole while drilling at 4107m to 4226m, increase mud weight from 104pcf to 105pcf.

Rigged up logging equipment and ran first full suite logging tool string as: DLL/SONIC/MSFL/CDR logging tool string, logged down from 3910m to TD at 4226m, logged up from 4226m to 100 m above shoe at 3810m and pulled out of hole to surface.

Ran second full suite logging tool string as: CNT/LDT/NGS/GR to 3910m, logged down from 3910m to TD at 4226m, logged up to 3900m and continued pull out of hole to surface. Due to low quality of second full suit result, replaced full suite logging tool string modules and again run in hole with same full suite logging tool string to 3910m, logged down from 3910m to TD at 4226m, logged up to 3900m and continued pulled out of hole to surface and rigged down logging equipment.

Held pre job safety meeting and ran 7" liner as following:

- 7" Float shoe 29ppf, C110, TPCQ (PPA Co.)
- 3 joints 7" liner 29ppf, S110, TPCQ
- 7" float collar 29ppf, C110, TPCQ (PPA Co.)
- 1 joint 7" liner 29ppf, S110, TPCQ
- 7" LDC 29ppf, C110, TPCQ (PPA Co)
- 1 joint 7" liner 29ppf, S110, TPCQ to 57m
- Circulated mud & function test float shoe & collar-ok.
- Continued run in hole with 13 joints 7"liner 29ppf, S110, TPCQ to 393m (totally 35 joints)
- Hydraulic liner hanger & acc. as following:
- 7" hydraulic liner hanger 29ppf, C-110, TPCQ (PPA Co)
- POB 29ppf, C-110 (PPA Co)
- Tie back sleeve (ID: 7-3/8", PPA Co) to 399m

Continued ran in hole with 7” liner string followed by 5" DPs from 399m to 4226m w/1 bph loss, tagged bottom at 4226m, pulled out to 0.5m off bottom at 4225.5m, circulated mud and dropped ball, set hydraulic liner hanger w/1800 psi and shear ball seat w/3000 psi, released setting tools.

Cement 7” liner in place as following:

- pumped 9bbl chemical wash, 7 bbl 85pcf mud push
- Mixed & pumped 220 sxs cement class "G" delijan blended with silica flour (SV:72 bbl, SW:118 pcf) • Dropped pump down plug, displaced with 266bbl 105pcf mud.
- Shearing after 223bbl with 2100psi, plug was not bumped, checked back flow-ok. Note: Surface sample of cement slurry, get 100% hard, after 6 hrs.

Ran in hole with 8-1/2” bit, tagged top of cement at 3680m and drilled out soft to medium cement from 3680m to 3826m.

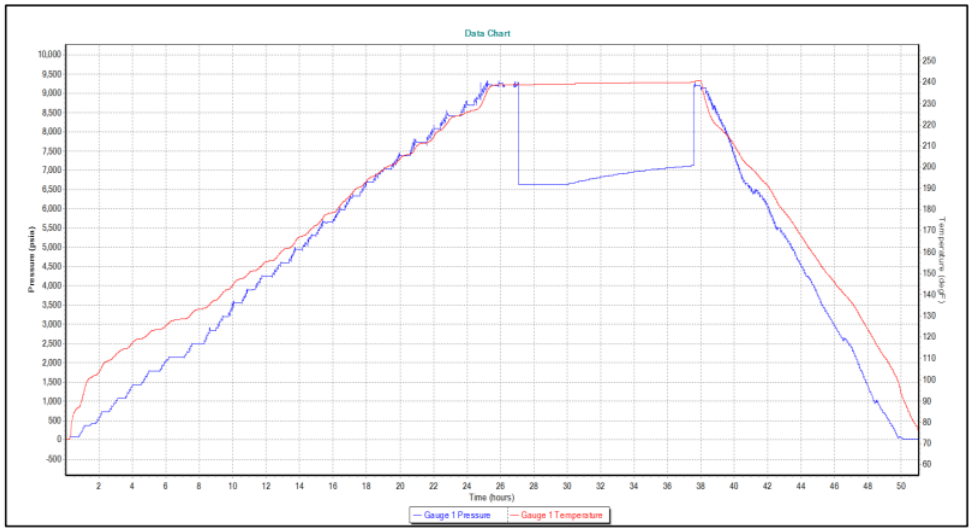
Note: Confirmed 7" liner lap at 3826m.

P/U 6-1/8" bit and ran in hole to 3830m, drilled out POB at 3830m, continued RIH to 4167m and resumed drill out hard cement to 6m above LDC at 4173m. Pulled out to surface.

Ran in hole with 6-1/8" used b. bit & 9-5/8" scraper to 7" liner lap 3826.5 m while scraped 9- 5/8" casing from 3650m to 3826.5m several times. Pulled out to surface.

Dry test: 1 st dry test, 7" liner: Rigged up dry test equipment and ran in hole to 3796m while filled up string with 170m 75pcf hi-vis pill and 3400m 85pcf mud as cushion.

L/U dry test line & surface equipment and tested same with 7500psi, set packer, opened hyd. spring valve for 3 hrs while observed no bubble on surface, closed DCIP valve and kept it closed for 6 hrs meanwhile observed annulus-static. Dropped bar, shear reverse sub, reverse circulated and got 208bbl cushion (198bbl 85pcf mud and 10bbl 75pcf hi-vis pill) on surface, continued circulation and balanced mud unset packer, flow check- well static. Pulled out of hole to surface and laid down dry test equipment.



NOTE: DCIP & hyd. spring sample properties; MW: 105pcf, PH: 10.5, Ca++: 1120 mg/l, Cl: 165000 mg/l.

Ran in hole with 6-1/8" B. bit & 7" scraper and scrap from 7" liner lap at 3826.5m to 4173m several times.

Rigged up logging equipment and ran in hole with VDL-SBT-WTC-GR-CCL logging tool string to top of cement at 4173m, logged up from 4173m to 3826.5m, continued pull out to surface and rigged down logging equipment.

Due to dry test result decided to tie back 7” liner. Tie-back:

P/U and M/U 7-1/4" polish mill & ran in hole to 7” liner lap at 3826.5m, polished liner lap to 3829.5m several times, pulled out of hole to surface and laid down 7-1/4" mill.

Rigged up liner running equipment and ran 7” liner tie back string as below:

- 7-3/8" seal nipple
- 3 joints 7" liner, S-110, TPCQ , 29ppf
- 7-3/4" LDC • 1 joint 7" liner, S-110, TPCQ, 29ppf to 40m, to check return circulated mud-ok.
- Continued ran in hole with 8-1/4" tie back sleeve (PPA) to 7” liner lap 3826.5m Sting in 7-3/8" seal nipple to 3827.5m, observed pressure up from 180psi to 580psi. Line up cement line and tested w/3500 psi-ok and performed cement job as below:
- Pumped 12bbl salt saturated as pre flush
- 7bbl 95pcf as spacer • 62.3sxs cement "G" Delijan (SV: 20bbl, SW: 118 pcf) as program. Dropped pump down plug, displaced cement slurry with 213bbl 105pcf mud, observed shearing with Max 1600psi, continued displacement with 1.8bbl 105pcf mud, bumped plug with 1300psi, sting in 7-3/8" seal nipple to 3829.5m, released setting tools, pulled out of hole and laid down setting tools.

P/U & M/U 8-1/2" bit and ran in hole to 3712m, drilled out medium to hard cement from 3712m to 7” liner lap at 3777m.

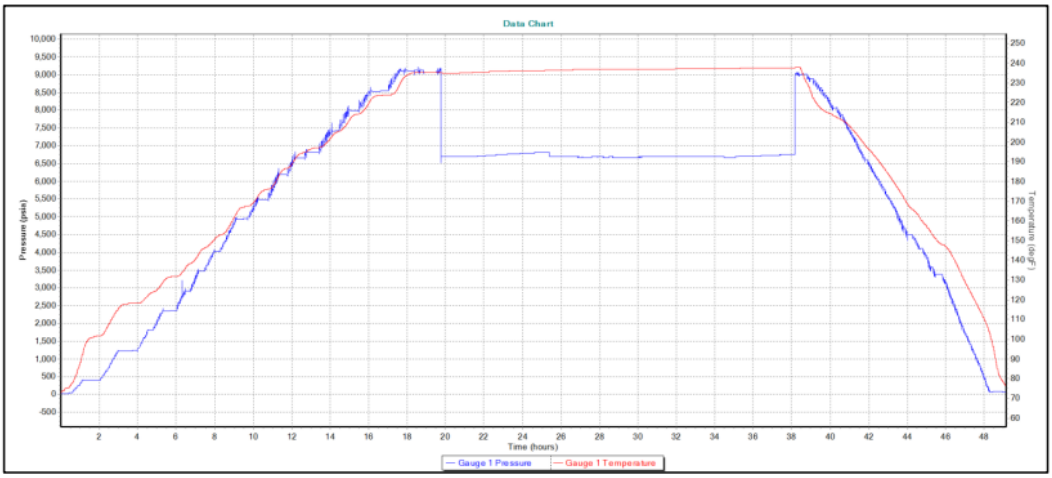
P/U & M/U 6-1/8" used bit to 3780.5 m, drilled out POB at 3780.5m, continued ran in hole to 3792m, drilled out plugs, LDC & hard cement from 3792m to 3826.5m, continued ran in hole to 4137m, pulled out to surface.

Dry test:

2nd dry test, 7” liner tie back:

Ran in hole with 6-1/8" bit and 9-5/8" scraper & ran in hole to 7" liner lap at 3777m while scraped from 3670m to 3777m several times. Pulled out to surface and laid down bit and scraper.

Ran in hole with 9-5/8" dry test tools while filled up string with 170m 75pcf hi-vis pill brine and 3350m 85pcf mud as cushion as per program to 3747m. Set packer at 3747m, 1st stage: opened hyd. spring valve for 2hrs, observed weak bubble. Closed DCIP for 4 hrs. 2nd stage: opened DCIP and kept it open for 3hrs, observed weak bubble, closed DCIP for 6hrs. Opened revers circulating sub, reveres circulated and got 199bbl cushion (189bbl 85pcf mud, 10bbl 75pcf hi-vis pill), continued circulation to balance mud, unset packer, flow check-well static. Pulled out and laid down dry test equipment.



Perforation and Completion

- **Perforation**

Ran in hole with 6-1/8" bit to top of cement at 4173m, circulated mud, pumped & displaced 33bbl 84pcf CaCl2 hi-vis pill (Length: 273m, from 4173m to 3900m), pulled out of hole to surface with NL-NG.

Loaded guns, performed pre job safety meeting, ran TCP/DST string while drifted DPs and filled up string with 79.5pcf CaCl2 as cushion to 4077m, Rigged up logging equipment and ran in hole with CCL/GR logging tool string to 3925m, logged up to surface and correlated depth, L/D logging equipment.

Set DST packer at 3954m, pressure tested DST packer by applying 400psi pressure in annulus for 15min-ok. Performed pre job safety meeting, N/U flow head, related lines and DST choke manifold, applied 1200psi to annulus, opened tubing tester valve, applied 3200psi pressure through string and step by step increased pressure to 4300psi, fired guns, released string pressure to zero, monitored tubing pressure: zero & annulus pressure: zero psi, applied 2800psi press in annulus, opened AOR valve, reversed circulated & got 217bbl 79.5 CaCl2 meanwhile observed no influx, continued reverse circulation to balance mud, N/D flow head and related lines, flow check-well static, pulled out of hole from 4077m to surface, L/D DST/TCP tools and checked TCP string, observed all bullets were fired.

To check perforated interval depth, R/U running logging equipment and ran CCL/GR logging tools to 4100m, logged up to 3700m and continued pulled out of hole to surface.

According to CCL/GR log result, perforation was done approximately 28m above planned interval, to double check again ran in hole with CCL/GR to 4100m and logged up to surface, the 2nd log showed same result as 1st log.

Planned perforation interval		Actual perforated interval	
Depth (m)	Formation	Depth (m)	Formation
3984m - 4005m	Fahliyan 1a and 1b	3956m – 3977m	Fahliyan top and 1a
4023m - 4026m	Fahliyan 1c	3995m – 3998m	Fahliyan 1b
4071m - 4077m	Fahliyan 3a	4043m – 4049m	Fahliyan 2a

Ran in hole with 6-1/8” E. mill to top of cement at 4173m, circulated mud and pulled out of hole to 4077m, spotted 20bbl 84.5pcf CaCl2 hi-vis pill, continued pulled out pooh from 4077m to surface.

To determine exact depth of perforated interval, rigged up logging equipment and ran in hole with MFC/CCL/GR logging tool string to 4100m, logged up from 4100m to 3930m and confirmed perforated interval as: 3956 to 3977m, 3995 to 3998m, 4043 to 4049m. Pulled out to surface and laid down logging equipment.

Ran in with 6” E. mill to 150m and circulated mud while waited on receive Detonator. After almost 180hrs NPT, received Detonator, rigged up logging equipment and ran in hole with 4-1/2" gun (w/o

bullet) & CCL (analog) as dummy run to 4090m, tool string held up, logged up to 7" liner lap at 3777m, pulled out to surface.

1st perforation run (WCP): Ran 1st perforation as: 4-1/2" gun (assembled with 99 No. bullet, phase: 60degree, length:6m) & CCL (Analog) log tool string to 4077m, perforated 6m, from 4071m to 4077m, pulled out to surface and observed all guns fired completely. Changed cable head.

2nd perforation run (WCP): Ran CCL/GR tool string to top of last perforation at 4071m, string held up, tried to pass several times-no success.

Pulled out to surface and changed cable head. Again ran in hole with 4-1/2" gun (assembled with 50 no bullet, phase: 60degree, length: 3m) & CCL log tool string to 4026m while correlated depth.

Perforated 3m from 4026m to 4023m, pulled out to surface and observed all guns fired completely.

3rd perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 99 no bullet, phase: 60degree, length: 6m) & CCL log tool string to 4030m while correlated & adjusted depth, checked and confirmed last perforated interval-ok, pulled out of hole to 4005m, fired gun & perforated 6m from 4005m to 3999m, pulled out to surface and observed all guns fired completely.

4th perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 50 no bullet, phase: 60degree, length:3m) & CCL log tool string to 3990.70m while correlated & adjusted depth, perforated 1m from 3999m to 3998m & 2m from 3995m to 3993m, pulled out to surface and observed all guns fired completely.

5th perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 99 no bullet, phase: 60degree, length: 6m) & CCL log tool string to 3984.7m while correlated & adjusted depth, perforated 6m from 3993m to 3987m, pulled out to surface and observed all guns fired completely.

6th perforation run (WCP): Ran in hole with 4-1/2" gun (assembled with 50 no bullet, phase: 60degree, length: 3m) & CCL log tool string to 3978.69 m while correlated & adjusted depth, perforated 3m from 3987m to 3984m, pulled out of hole to surface and observed all guns fired completely.

Changed cable head, ran CCL/GR logging tools to 3984m, string held up, confirmed top of perforated interval, logged up to 3700m, pulled to surface and rigged down logging equipment.

Ran in hole with 6" E. mill from 1700m to 3984m, took weight, continued ran in hole with rotation & pumping from 3984m to 3994m, continued run in hole with pumping to bottom of perforated interval at 4077m, pumped & displaced 12bbl hi-vis pill and 21bbls 84pcf CaCl2 (277m, from 4077m to 3800m)

After 411hrs wait on receive Wixiqin company completion running crew and equipment, ran in hole with 6" E. mill to 3950m and displaced 105pcf mud in hole with 104pcf CaBr2 as completion fluid w/ NL-NG as following: Pumped 50bbl 105pcf hi-vis pill, 30bbl 85pcf non-solid CaCl2 as spacer, 820bbl 104pcf CabBr2, got 5bbl contaminate, observed well for 5hrs-static.

Run Completion string:

- 1 st Completion running:

Retrieved wear bushing, held pre job safety meeting and ran completion tally as following:

- 3-1/2" mule shoe 10.2ppf, HSM-1, L-80.13Cr
- 1 joint 3-1/2" tubing HSM-1, 10.2ppf, L-80
- 3-1/2" bottom No-Go 10.2ppf, HSM-1, L-80.13Cr (w/2.562" xn plug & prong inside it)
- 3-1/2" pup joint
- 1 joint 3-1/2" tubing
- 7" Permanent production packer model WSDP
- 3-1/2" Anchor seal, 1 joint 3-1/2" pup joint
- 3-1/2" tubing, 1 joint 3-1/2" pup joint, 3-1/2" SSD (in open position w/ 2.750" x plug & prong inside it), 1 joint 3-1/2" pup joint
- 1 joint 3-1/2" tubing, 1 joint 3-1/2"pup joint, 3-1/2" travelling joint (in full open position w/10 ft stroke), 1 joint 3-1/2" pup joint
- 1 joint 3-1/2" tubing, 1 joint 3-1/2" pup joint, 3-1/2" CIV 1, 1 joint 3-1/2" pup joint

• 1 joint 3-1/2" tubing,1 joint 3-1/2" pup joint, 3-1/2" CIV 2, 1 joint 3-1/2" pup joint, 1 joint 3-1/2" tubing. Pressure test completion string w/8000 psi-ok, connected 3/8" CIL to both CIV no1 & CIV no2 and pressure tested same to 4000psi for 15min-ok.

• Continued ran in hole with 17 joint tubing, 3-1/2", 10.2ppf, L-80, HSM-1

• XOS 4 1/2" HSM-1 box X 3 1/2" HSM-1 pin

• 1 joint 4-1/2" tubing, 18.9ppf, L-80, HSM-1 while hold 1000psi pressure inside CIL & filled up string with CaBr2 as completion fluid each 10 joint. Pressure tested completion string at 2530m w/8000 psi for 15 min-ok.

• Rigged up slick line, ran 2.750" gauge ring to above SSD, Laid down slick line equipment

• 133 joint 4 1/2" tubing, 18.9ppf, HSM-1, L-80 while fill up string with CaBr2 as completion fluid each 10 joints with 0-1 bph loss.

Pressure tested completion string at 1480m with 8000psi -no success, rigged up slick line, ran 2" SB-pulling to top of SSD and pushed down prong, pulled out slick line to surface, tested with 8000psi -no success.

Retrieved 2.750" prong & plug from SSD, changed O-ring and packing, again set prong & plug in SSD, tested with 8000psi several times-no success, pulled out completion string to 1200m while pressure tested completion string at 1340m & 1200m with 8000psi- no success, continued pull out to 1050m, pressure tested completion string with 8000psi for 15 min-ok.

• Ran in hole with completion string to 1140m with 0-1 bph loss, pressure tested completion string with 8000psi-no success. Pulled out of hole from 1140m to 1113m, laid down 1 stand 4-1/2" tubing, tested completion string with 8000psi for 15 min-ok.

• Ran in hole to 3856m while tested completion string at 1282m, 1450m, 2135m, 2790m, 3443m and 3865m with 8000psi for 15min-ok.

• Continued ran completion string to 3880m as:

• 1 joint 4-1/2" pup joint, 4-1/2" flow coupling 4ft, 4-1/2" x 3.437" TRSSV, 10k, weiqixin, 4-1/2" flow coupling 6ft, 1 joint 4-1/2" pup joint, 1 joint 4-1/2" tubing, 18.9ppf, l-80, HSM-1. Connected

1/4" control line to TRSSV, tested it with 8000psi-ok, pressure tested completion string with 8000psi for 15min-ok, closed TRSSV and released pressure above TRSSV to 500psi, as inflow test held pressure for 15 min-ok, increased pressure above TRSSV to 6000psi and opened TRSSV by apply 3750psi as equalizing test,.

- Continued ran completion string to 3930m with 5 joints 4-1/2" tubing 18.9ppf, L80 , 4-1/2" fluted swage, double pin Vam top x HSM-1, MSP 11 x 4-1/2" tubing hanger, 10k ACME box up x Vam top box down, tested completion string with 8000psi for 15min-ok. M/U 4-1/2" landing joint, connected 1/4" control line to TRSSV and 2 no 3/8" CIL to hanger, tested 1/4" control line with 8000psi and tested 3/8" CIL with 1000psi- ok, ran completion string to 3933m, landed tubing hanger to THS, to space out pulled out to 3931.5m.

Rigged up slick line equipment, installed bop and slick line lubricator, tested it with 4000psiok and ran with slick line; 1 st run: retrieved prong of 2.750" plug from SSD while circulated cross well head with 0-1bph loss.

As 2nd run, retrieved 2.750" plug "X" type from SSD

As 3rd run, closed SSD sleeve by shifting tools, rigged down slick line equipment, line up test truck, step by step increased completion string pressure to 3500psi, set packer at 3906m, pull test packer with 15klb-ok, landed tubing hanger in THS, closed all tie down screws, pressure tested completion string with 8000psi, observed pressure dropped and also return from annulus, pressure tested annulus with 2000psi observed pressure dropped and also return from completion string.

As 4th run, ran with shifting tools to check SSD is closed, pulled out to surface.

As 5th run, ran with 2.750" selective tester tools to above SSD, to check string. Pressure tested completion string with 8000psi for 15min-ok. Rigged up slick line equipment.

As 6th run, ran with GS pulling, retrieved 2.750" selective tester tools of SSD.

As 7th run, ran with 2" SB-pulling to top of bottom No-go landing nipple, pushed down prong and tested completion string with 2000psi, observed return from annulus.

As 8th run, ran with shifting tools, opened SSD, circulated from string and got return from annulus.

As 9th run, ran with shifting tools and closed SSD, tested completion string with 3000psi, observed return from annulus.

As 10th run, ran with 2.562" SB-pulling, retrieved prong & plug from bottom No-go in two run.

As 11th run, ran with 2.562" selective tester tools to No-go landing nipple while pressurized inside completion string to 7000psi, observed return from annulus.

As 13th run, ran in hole with GS pulling and retrieved 2.562" selective tester tools and changed redress kit. again ran with 2.562" selective tester tools to top of No-go, pressure tested inside completion string with 7000psi, observed return from annulus.

Ran PLT logging equipment, ran with PLT logging tool string as; FCF, GCT, PWH, HSTC to 3918m and line up to 3840m while pressurized string in two stages, first stage applied 7000psi inside completion string and got return from annulus, as second stage applied 2600psi and got return from completion string, PLT result: recorded leakage at 3-1/2" anchor seal assembly and 7" permanent production packer. Pulled out to surface, rigged down PLT logging equipment, r/u slick line equipment.

As 14th run, ran with GS-pulling, retrieved 2.562" selective tester tools from No-go landing nipple, set 2.562" plug & prong in No-go landing nipple in two stage

As 16th run: ran with 2.750" selective tester in SSD. Picked up 1.5m completion string to full stork traveling joint, pulled string with maximum 25klb over pull, slack off weight and applied 2000psi pressure inside string, ran string to half close traveling joint, applied 50klb weight on anchor seal inside packer, landed tubing hanger and closed tie down screws, released string pressure to zero.

Rigged up slick line equipment and as 17th run, ran with SB-pulling tools, retrieved 2.750" selective tester tools from SSD, tested completion string with 2000psi, observed return from annulus, tested completion annulus with 2000psi-ok.

As 18th run, set protection sleeve in TRSSV, tested completion string, observed return from annulus, tested completion annulus with 2000psi- ok.

As 19th run, ran with 2.750" selective tester tools in SSD. Picked up 1.5m comp string to full open travelling joint, applied 5klb over pull, rotated string 11 round to right, unlatch anchor seal from

packer, half close travelling joint and landed tubing hanger in THS while applied 50klb weight on anchor seal & 2000psi pressure in string to latch it in 7" packer again, observed increased pressure to 2300psi, tight tie down screw, released string pressure to zero, pull test completion string with 20klb-ok. Pressure tested from annulus with 2000psi-ok, rigged down slick line equipment, ran with 2.570" SB-puling, retrieved 2.570" selective tester tools from SSD, pressure tested completion string with 2000psi, observed return from annulus, retrieved prong & plug from No-go landing nipple in two, rigged down slick line equipment, opened t.ie down screw, pulled out 1.5m, rotated string 11 round to right, unlatch 3-1/2" anchor seal from packer. Pull out completion string to surface and laid down same.

Ran in hole with 6-1/8" b. bit to 6m above packer at 3900m and circulated cross wellhead, while wait on receive completion accessories with 1bph loss, to check packer depth ran in hole to 3906m, confirmed packer depth at 3906m, circulated bottoms up with 0-1bph loss, continued wait on receive completion equipment while circulated cross well head and pumped through string each 2hrs, pulled out from 3906m to surface.

- 2 nd Completion running:

Rigged up completion running equipment, ran comp string 3-1/2" x 4-1/2" as below:

- 4-1/2" anchor seal, 3-1/2" pup joint, 3-1/2" bottom no-go landing nipple XN 2.56" without plug, 1 joint 3-1/2" pup joint, 1 joint 3-1/2" tubing, 10.2ppf, HSM-1, 1 joint 3-1/2" pup joint, XOS 4-1/2" HSM-1 box x 3-1/2" HSM-1 pin
- 4-1/2" mill out extension, 7" permanent production packer (hs-6 model, weiqixin)
- 4-1/2" anchor seal , 1 joint 3-1/2" pup joint,1 joint tubing, 10.2ppf, HSM-1
- 1 joint 3-1/2" pup joint, 3-1/2" SSD (open position) & 2.750" with check valve
- 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1
- 1 joint 3-1/2" pup joint, expansion joint 3-1/2", HSM-1, L-80, 10 stroke (full open, weiqixin)
- 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1

• 1 joint 3-1/2" pup joint, 3 1/2" CIV 1 tubing retrievable, HSM-1, 10.2ppf with checked valve, 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1, 1 joint 3- 1/2" pup joint, 3 1/2" CIV 2 tubing retrievable, HSM-1, 10.2ppf with check valve

• 1 joint 3-1/2" pup joint, 1 joint tubing, 10.2ppf, HSM-1 to 86m Line up test truck, pressure tested line with 8000psi-ok, pressure tested completion string with 8000psi for 15min-ok, connected 2 no 3/8" CIL to CIV and tested them with 4000psi for 15min, ok, continued run completion string with 17 jts 3-1/2" tubing, 10.2ppf, HSM-1

• XOS 3-1/2" HSM-1 pin x 4-1/2" HSM-1 box, 1 joint 4-1/2" tubing, L-80, 18.9ppf, HSM-1 to 247m. Pressure tested completion string with 8000psi for 15 min-ok, rigged up slick line equipment, ran 2.750" gauge ring to top of SSD-ok, rigged up slick line equipment, continued run completion string from 247m to 950m.

• Continued run completion string from 950m to 3837m with 4-1/2" tubing, L-80, 18.9ppf, HSM-1 meanwhile passed 3.5" drift through all 4-1/2" tubing, pressure tested completion string with 8000psi at 1592m, 2575m and 3837m-ok.

Rigged up slick line, run 2.8" gauge ring to top of SSD, rigged up slick line equipment, ran with 3" SB pulling to top of SSD, tried to catch 2.750" check valve several times, catch and retrieved check valve from SSD after several attempts, reverse circulated with completion fluid, installed new 2.750" check valve inside SSD, rigged down slick line equipment, pressure tested completion string with 8000psi- not ok, pulled out completion string to 3640m, pressure tested completion string with 8000psi- not ok, rigged up slick line equipment, ran 3" SB pulling to top of SSD, tried to engage with 2.750" check valve several times-no success, pulled out to surface, checked tools and ran again with 3" SB pulling to top SSD, catch 2.750" check valve & tried to remove it several times-no success, pulled out to surface.

As 3rd run, retrieved check valve from SSD. Pulled out with completion string from 3640m to 2240m.

1st and 2nd run, tried to install 2.750" plug in SSD several times-no success, pulled out with slick line to surface and changed plug.

3rd run, installed 2.750" plug in SSD,

4th run, installed prong inside plug, rigged down slick line, pressure tested completion string with 8000psi, observed pressure dropped, rigged up slick line, 1st run, tried to engage prong by SB pulling-no success, 2nd run, retrieved prong from SSD plug, 3rd run, retrieved 2.750" plug from SSD, rigged down slick line equipment.

Pulled out with completion string from 2240m to 946m, rigged up slick line equipment, 1st run, installed 2.750" check valve inside SSD, tested completion string with 8000psi-not ok, 2nd run, retrieved 2.750" check valve from SSD, rigged down slick line equipment, due to lack of enough completion fluid on surface, stopped pull out and circulated mud cross well head while wait on receive completion fluid with 0-1bph loss.

Received 68bbl CaBr2, pulled out with completion string from 946m to 76m, rigged up slick line equipment, installed 2.750" check valve inside SSD, pressure tested completion string at 76m with 8000psi, observed 250psi dropped in pressure in 20min, continued pull out to surface while pressure test and visually check completion string on travelling joint and SSD and CIV with 8000psi-ok, tested CIL, observed rupture disks were broken.

- 3 rd completion running:

Connected 2no 3/8" CIL to 1" CIV and pressure tested same with 4000psi for 15min-ok, ran completion string as following:

• 5 stand 3-1/2" tubing, 2 joints 3-1/2" tubing, 10.2ppf, L-80, HSM-1, XOS 4-1/2" HSM-1 box * 3-1/2" HSM-1 pin, 1 joint 4-1/2" tubing, 7 stand 4-1/2" tubing, 18.9ppf, HSM-1 with 0-1bph loss.

• 4-1/2" tubing 18.9ppf, L-80, HSM-1, total: 71 stand, meanwhile ran 2.8" gauge ring by slick line at 640m to top of SSD & pressure tested completion string at 640m & 2240m with 8000psi for 15min-ok.

• continued ran completion string from 2240m to 3837m with 57 stand 4-1/2" tubing 18.9ppf, l-80, HSM-1, total:128 stand, pressure test completion string with 8000psi for 15min-ok. Rigged up slick line equipment, ran 2.8" gauge ring to top of SSD, rigged down slick line equipment, continued ran in completion string to 3874m as below:

• 1 joint 4-1/2" pup joint, 4 ft flow coupling, 4-1/2" * 3.437" TRSSV-10k 6ft flow coupling, 1 joint 4-1/2" pup joint, 1 stand 4-1/2" tubing, connected 1/4" control line to TRSSV & tested it with 8000psi

for 15min-ok with 0-1bph loss. Pressure tested comp string at 3874m with 8000psi for 15min-ok, as inflow test closed TRSSV, released pressure above TRSSV from 8000 to 300psi, monitored pressure for any charge for 15min-ok.

- Continued ran completion string to 3894m with 5 joints 4-1/2" tubing, 4-1/2" fluted swage adapter, MSP tubing hanger 11" * 4 1/2" 10k, pressure tested completion string with 8000psi for 15min-ok.

- Connected 1/4" control line to hanger, tested same with 8000psi-ok, connected 2 no 3/8" CIL to hanger, continued ran completion string to 3906.3m. Sting anchor seal sting in old packer with 0-3klbs, placed tubing hanger 1.5m above THS, set protect sleeve in TRSSV landing nipple, removed check valve from SSD, tried to close SSD in two attempts-no success. Changed shifting tools and tried to close SSD-no success, after 4 attempts, pined shifting tools with steel shear pin, continued ran in hole with same and closed SSD, ran 2" gauge ring as dummy run to No-go landing nipple 3902m, string held up, tried to circulate, pressure increased to 300psi, tried to revers circulate, pressure increased to 300psi- no success, pulled out to surface and rigged down slick line.

Applied 500psi pressure inside string, tried to circulation-no success, applied 700psi pressure in annulus and tried to reverse circulation- no success, released pressure, tried to pull out, observed 30klb overpull, to land tubing hanger in THS, applied 50klbs weight, rigged up slick line and ran 2.3" impression block to 3900m, worked on same and pulled out to surface, no mark on impression block, again ran 2.3" impression block to 3900m, worked on same and pulled out to surface, no mark on impression block.

Worked on completion string upwards with maximum 30klb over pull several-no success, rigged up slick line equipment and ran 1.8" sand bailer to 3905 , worked on bailer and pulled out to surface, recovered no sample, ran with shifting tools to SSD and closed same, pulled out to surface, pressure tested completion string with 500psi-ok.

While waited on Weiqixin Company due to stopped operation by weiqixin crew, unlatched anchor seal from packer by applying 10klb over pull and rotated 12 round to right while held 300psi pressure inside completion string, pulled out to 0.5m above packer while circulated with NL-NG, latched anchor seal to packer by applying 10klb weight, worked on completion string upwards with maximum 30klb over pull several times- no success, unlatched again anchor seal from packer and pulled out completion string while circulated cross well head with NL-NG.

Ran in hole with 3-1/2" anchor seal to 3890m, latched anchor seal inside packer and tried to recover packer by applying maximum 90klb over pull several times-no success, anchor seal got unlatch, tried to latch several times-no success Pulled out from 3890m to surface, laid down all tubing**.**

• **Milling:**

Ran in hole with 6-1/8" metal muncher junk mill to 5m above top of packer at 3885m, circulated and displaced CaBr2 with 104pcf kcl-polymer mud, continued ran in hole from 3885m to 3890m, tagged top of packer at 3890m. Commenced milling packer from 3890m to 3890.8m, recovered 44lb junk and element rubber of packer from shaker, pulled out from 3890.8m to surface, ran in hole with 2-1/4"*4" taper tap and to top of packer 3891.8m, tap inside packer with 2-4klb weight, catch fish and pulled out with 20-40klb drag to surface, completely recovered tail of packer as : 3-1/2" anchor seal, 3-1/2" pup joint, 3-1/2" bottom no-go, 3-1/2" pup joint, 3-1/2" tubing 10.2ppf, L-80, HSM-1, XOS 3-1/2" * 4-1/2" HSM-1, 4-1/2" mill out extension.

Ran in hole with new 6-1/8" metal muncher junk mill to 3908m, washed down from 3908m to 3910m, tagged top of packer at 3910 m, commenced milling packer from 3910m to 3910.65m, recovered 30lb junk & rubber element of packer from shakers, pulled out to surface.

Ran in hole with 2-1/4" x 4" taper tap to top of packer at 3910.65m, tap inside packer with 2-4klb weight, worked on same and pulled to surface, not recovered fish, ran in hole with same 2-1/4" x 4" taper tap to top of packer at 3910.65m, circulated mud with high flow rate, worked on catch fish with no success. Pulled out to surface and laid down taper tap. Ran in hole with 6-1/8"metal muncher junk mill to top of packer at 3910.65m, commenced milling from 3910.65m to 3910.7m, fish slipped in hole, pushed fish down to 3950.50m and resumed milling to 3951.4m, pulled out to surface.

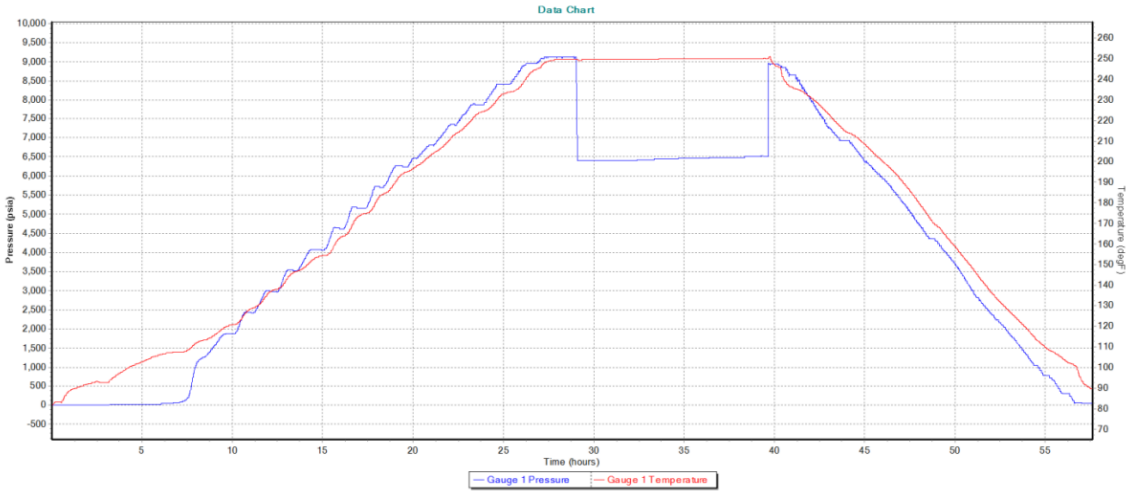
Ran in hole with 2-1/4" x 4" taper tap to top of fish at 3951.4m, break circulation, tap inside packer and worked on catch fish, pulled out from 3951.4m to surface, completely recovered tail of packer as: 3-1/2" mule shoe, 3-1/2" tubing 10.2ppf, L-80, HSM-1, 3-1/2" bottom nogo, 3-1/2" pup joint, 3-1/2" tubing 10.2ppf, L-80, HSM-1, 3-1/2" pup joint, XOS 3-1/2" * 4- 1/2" HSM-1, 4-1/2" mill out extension.

Plug and Suspension:

Ran 6-1/8" b. bit and 7" scraper to 3900m while scraped from 3780m to 3900m, circulated mud while added 0.5 % vol. biocide and 1% vol. brine to mud system and pulled out to surface.

Ran in hole with 7" hydro mechanical bridge plug to 3875m, dropped ball, circulated mud, tried to set 7" bridge plug by applying 3000psi-no success, line up test truck, tried to set hydro mechanical bridge plug by applying pressure to 4000psi-no success, reverse circulated for 15min in 2 stage and several times jerked string, applied 4000psi inside string, set 7" bridge plug at 3875m successfully, released press, checked string upward with 35klbs over pull and slack of 10klb weight-ok, released running tools, pulled out to 3760m, closed top rams & tested bridge plug with 2000psi-ok, continued pull out to surface.

Ran in hole with 7" dry test tools to 3625m while filled up string with 210bbl w/75 pcf non solid brine as water cushion, set packer at 3845m, opened hydro spring and kept open for 3hrs, observed no bubble, closed DCIP valve for 6hrs, dropped bar, shear revers sub and reveres circulated, got water cushion on surface and observed no influx effect, continued circulation to balance mud at 104 pcf, unset packer, pulled out to surface and recovered chart as below**:**



1st cement plug:

- Ran in hole with 6-1/8" center cut to 3700m, set first cement plug as below:
- Pumped 15bbl spacer.

- Pumped 300 sxs blended silica flour cement "G" Kerman, slurry weight: 118pcf and slurry volume: 35bbl.
- Pumped 3.7bbl behind spacer and displaced with 200bbl mud as balance plug.

Pulled out from 3700m to 3520m, flush DPs, continued pull out to 3350m, circulated bottoms up, and waited on cement to get hard, tagged hard cement with 470gpm and 15klbs-ok at 3513m, pulled out from 3513m to 3500m, pressure tested against cement plug with 2000psiok, pulled out from 3500m to 1480m.

2nd cement plug:

At 1480m, break circulation, line up cement line and tested same with 4000psi-ok, set 2nd cement plug as below:

- Pumped 15bbl 97pcf ahead spacer.
- Pumped 300sxs blended silica flour cement class "G", slurry volume: 118pcf, slurry weight: 35bbl.
- Pumped 3.7bbl behind spacer and displaced with 72bbl mud as balance plug. Pulled out from 1480m to 1300m, flush DPs, continued pulled out to 1000m, circulated bottoms up, waited on cement to get hard, continued ran in hole from 1200m to 1317m, tagged cement with 400gpm &15klb, circulated bottoms up, pulled out to surface, laid down drill pipes and string.

Landed tubing hanger in THS and tight tie down screws, back off 5" landing joint, installed 4" two way in tubing hanger and tested it with 1000 psi-ok.

N/D 13- 5/8" BOP stack (10k) & disconnect outlet valves, installed 11" * 4-1/16" bottom master valve (10k, msp), tested tubing hanger & o-ring & bottom master valve connection & metal seal with 4000psi-ok, removed 4" two way from tubing hanger, pressure tested bottom master valve with 2000 psi-ok, installed 4" one way in tubing hanger, tested it with 2000psi-ok, installed blind flang on bottom master valve and tested it with 2000psi-ok.

Secured well and released rig 304 on 96/06/12 at 24:00hrs.

Re-entry section:

Spudded the rig on 1396.11.03 at 06:00, retrieved 11" * 4-1/16" 10k MSP tubing hanger and checked same-ok, observed the well-static, installed 11" wear bushing, ran in hole with 8- 1/2" bit to 1315m, tagged 1st cement plug at 1315m, circulated mud and balanced mud weight in/out at 104pcf, observed the well- static, drilled out medium cement from 1315m to 1350m and medium to hard cement plug from 1350m to 1480m with NL-NG, circulated bottom's up to clean the well, continued run in hole from 1480m to 3513m, tagged 2nd cement plug at 3513m, circulated to balance mud weight at 104 pcf, observed the wellstatic, drilled out soft to medium cement from 3513m to 3665m and medium cement from 3665m to bottom at 3700m, observed the well-static, ran in hole from 3700m to 7" liner lap at 3777m, circulated mud to balance mud weight in/out at 105pcf, observed the well- static, pulled out from 3777m to surface.

Ran in hole with 6-1/8" new junk mill and tagged 7" bridge plug at 3863.7m, pumped 30bbl hi-vis pill, circulated mud and milled 7" bridge plug from 3863.7 to 3864.08m, due to no progress pulled out from 3864.08m to surface, laid down 80% used 6-1/8" junk mill, ran in hole with 6-1/8" new junk mill to 3864.08m and tagged 7" bridge plug, resumed milling 7" bridge plug from 3864.08 to 3864.38m, 7" bridge plug dropped in hole, observed the wellstatic, ran in hole to top of cement plug at 4173m while pushed down 7" bridge plug with pumping and rotation, circulated mud to balance in/out at 105pcf, spotted 30bbl CaCl2 with NL-NG, pulled out from 4173m to surface and laid down 70% used 6-1/8" junk mill.

Ran in hole with 4-1/8" used junk mill to top of cement plug at 4100m while scraped 7" liner from 3777m to 3930m and 9-5/8" casing from surface to 3777m, circulated bottom's up with NL-NG, pulled out from 4100m to surface and laid down 4-1/8" used junk mill.

Tested BOP and ran in hole with 4-1/8" used junk mill to 3750m and waited on receive CaBr2 while slow circulated through string with NL-NG, pulled out from 3750m to 1400m, rigged up running tubing equipment and stand back 168 joints 4-1/2" tubing while waited on receive CaBr2, pulled out to surface.

Ran in hole with 4-1/8" junk mill to 4173m, circulated bottoms up and spotted 30bbl CaCl2, pulled out from 4173m to 3750m, waited on receive CaBr2 while slow circulated with NL-NG, ran in hole from 3900m to 4100m, line up pump truck and displaced 105pcf mud with 106pcf CaBr2, pumped

30bbls 85pcf hi-vis pill CaCl2as spacer, pumped 840bbls 106pcf CaBr2 and got 10bbls contaminated with NL-NG, observed the well for 4hrs-static, pulled out from 4100m to Surface while laid down DPs.

1st Attempt:

Pressure tested BOP, rigged up OIDS tubing running equipment, held PJSM and ran 3-1/2" x 4-1/2" completion string as below:

- 3-1/2" mule shoe 10.2ppf, L-80, HSM-1, 13Cr
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1
- 3-1/2" bottom No-go 10.2ppf, L-80, HSM-1, 13Cr with 2.562" seal bore type XN
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- XOS 3-1/2" HSM-1 pin X 4-1/2" HSM-1 box
- 4-1/2" mill out extension 18.9ppf, HSM-1
- 7" permanent production packer model WSDP, 3-1/2" Anchor seal, 3-1/2" pup joint 10.2ppf, L80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" SSD, in open position with 2.750" check valve inside it
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" expansion joint (10 ft stroke full open) 10.2ppf, L-80, HSM-1, 3-1/2" pup joint 10.2ppf, L-80, HSM-1

- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 3-1/2" CIV 1 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 3-1/2" CIV 2, 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1 to 93m

Pressure tested completion string with 8000psi for 15min-ok, rigged up slick line, ran 2.8" gauge ring to above check valve inside SSD, rigged down slick line equipment, connected 3/8" CIL to both CIV 1 and CIV 2, press tested CIL with 4000psi for 15min- ok, continued run completion string from 93m to 250m as:

- 13 joints 3-1/2" tubing 10.2ppf, L-80, HSM-1 • XOS 4-1/2" HSM-1 box * 3-1/2" HSM-1 pin
- XOS 4-1/2" TPG-2 box * 4-1/2" HSM-1 pin
- 1 stand 4-1/2" tubing 18.9ppf, L-80, TPG-2

while kept 1500psi pressure inside of CIL Pressure tested completion string at 250m with 8000psi for 15min-ok, rigged up slick line, ran 2.8" gauge ring to above check valve inside SSD, rigged down slick line equipment, continued run completion string to 3800m with 4-1/2" tubing 18.9ppf, L-80, TPG-2 while used cross coupling each joint with NL-NG and while pressure tested completion string at 2422m, 3000m and 3800m with 8000psi for 15min-ok.

Continued run in hole as:

- XOS 4-1/2" TPG-2 pin * 4-1/2" HSM-1 box 18.9ppf, L-80
- 4-1/2" pup joint 18.9ppf, L-80. hsm-1
- 4-1/2" flow coupling 18.9ppf L-80, super 13Cr HSM-1

- 4-1/2" TRSSV 18.9ppf, L-80, super 13 Cr, HSM-1 (profile: 3.437")
- 4-1/2" flow coupling 18.9ppf, L-80, super 13Cr, HSM-1
- 4-1/2" pup joint 18.9ppf, L-80, HSM-1
- XOS 4-1/2" HSM-1 pin * 4-1/2" TPG-2 box 18.9ppf, L-80 to 3820m Tested completion string with 8000psi for 15min- ok meanwhile inflow tested TRSSV-ok, continued run completion string as: • 3 joints 4-1/2" tubing 18.9ppf, L-80, TPG-2
- XOS 4-1/2" TPG-2 pin * 4-1/2" HSM-1 box 18.9ppf, L-80
- 4-1/2" fluted swage adapter 18.9ppf, L-80, Super 13Cr V. top * HSM-1
- 11* 4-1/2" tubing hanger 10k MSP Drilex to 3860m.

Tested completion string with 8000psi -not ok, tested again with 8000psi- not ok, laid down 11* 4-1/2" tubing hanger and 4-1/2" fluted swage adapter 18.9ppf, L-80, Super 13Cr V. top * HSM-1 to 3857m, tested completion string with 8000psi –not ok.

Pulled out from 3857m to 3820m and stand back 1 stand 4-1/2" tubing 18.9ppf, L-80, TPG-2, tested completion string with 8000psi- not ok, rigged up slick line, set 3.437" protection sleeve in TRSSV, ran with SB pulling and tried to retrieve 2.750" check valve inside SSD at several time - not succeed and tried to reverse circulated by applying max 500psi -not succeed. Ran 2.750" impression block to SSD depth at 3787m and worked on same, pulled out to surface no mark observed on impression block, rigged down slick line.

Ran completion string as:

- 3 joints 4-1/2" tubing 18.9ppf, L-80, TPG-2
- XOS 4-1/2" TPG-2 pin * 4-1/2" HSM-1 box 18.9ppf, L-80
- 4-1/2" fluted swage adapter 18.9ppf, L-80, Super 13Cr V. top * HSM-1
- 11* 4-1/2" 10k MSP Drilex tubing hanger
- XOS 4-1/2" Acme pin * 4-1/2" N. Vam box 18.9ppf, L-80

• 4-1/2" tubing 18.9pf, L-80, N. Vam landing joint to 3870m Landed tubing hanger inside 13-5/8" * 11" tubing head spool, pulled out to 3868.5m (1.5 m space out packer position). Nipple up 4-1/16" G. flange, rigged up coil tubing, pressure tested BOP and stuffing box with 5000psi for 20min- ok, tensile tested with 20klb for 20min- ok, flush coil tubing, ran with 1- 1/2" coil tubing to SSD depth 3831m while circulated with max rate 0.7bbl/min, washed and cleaned above check valve several times, pulled from 3831m to surface, rigged down coil tubing with NL-NG, rigged down slick line, ran with SB pulling and retrieved 2.750" check valve inside SSD, pulled out to surface, rigged down slick line, set 4-1/16" G. flange, rigged up coil tubing, pressure tested BOP and stuffing box with 5000psi for 20min- ok, tensile tested with 20klb for 20 min- ok, flush coil tubing. Ran in hole with 1-1/2" coil tubing to mule shoe at 3870m while pumping, washed & cleaned through string, circulated bottoms up meanwhile added o2 scavenger as per program to completion fluid system with NL-NG. Pulled out from 3870m to surface, flush coil tubing, rigged down coil tubing.

Rigged up slick line, ran with 2.750" check valve to SSD depth at 3831m, set 2.750" check valve in SSD, pulled out to surface, tested completion string with 8000psi- not succeed, ran in hole with SB pulling and retrieved 2.750" check valve, pulled out to surface, ran with 2.750" plug to SSD depth 3831m and set plug in SSD, pulled out to surface, ran with 2.750" prong to 3831m, set prong in plug, pulled out to surface, rigged down slick line, tested completion string with 8000psi- not ok. To check string pulled out comp string from 3870m to 3805m while tested completion string at 3820m and 3805m with 8000psi- not ok, rigged up slick line, ran in hole with SB pulling to SSD depth, tried to retrieved prong and prong in two run, rigged down slick line, pulled out completion string from 3805m to 2478m, rigged up slick line and ran in hole with 2.750" check valve to SSD depth, set check valve, pulled out to surface and rigged down slick line, pressure tested completion string with 8000psi- not ok, rigged up slickline, ran in hole with SB pulling to SSD depth, retrieved check valve, pulled out to surface, rigged down slick line, pulled out of hole from 2475m to 84m, tested completion string with 8000psi- not ok, continued pull out to 58m, pressure tested completion string with 8000psi- not ok, observed leakage from SSD, laid down 3-1/2" expansion joint and 3-1/2" SSD, continued pull out to surface, observed Packer rubber element deformed and damaged, laid down the same, checked bottom no-go profile- ok.

2nd Attempt:

Ran 3-1/2" * 4-1/2" Completion string as following:

- 3-1/2" mule shoe 10.2ppf, L-80, HSM-1, 13Cr
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1
- 3-1/2" bottom no-go 10.2ppf, L-80, HSM-1, 13Cr with seal bore 2.562" type XN
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint tubing 3-1/2" 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- XOS 3-1/2" HSM-1 * 4-1/2" TPG-2
- 4-1/2" mill out extension 18.9ppf, TPG-2
- 7" permanent production packer model WSDP, 3-1/2" Anchor seal, 3-1/2" pup joint 10.2ppf, L80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1 to 36 m

Rigged up slick line, ran run 2.800" gauge ring to top of bottom no-go, continued run completion string as:

- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" SSD (in close position with 2.750" check valve inside it, 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" expansion joint (10 ft stroke, full open) 10.2ppf, L-80, HSM-1, 3-1/2"pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1 to 68m Line up test truck, tested completion string at 40m and 68m with 8000psi for 15min-ok, continued run completion string as:
- 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" CIV1 10.2ppf, L-80, HSM-1, 3-1/2" pup joint 10.2ppf, L-80, HSM-1
- 1 joint 3-1/2" tubing 10.2ppf, L-80, HSM-1

• 3-1/2" pup joint 10.2ppf, L-80, HSM-1, 3-1/2" CIV2, 3-1/2" pup joint 10.2ppf, L-80, HSM-1 to 84m Tested completion string with 8000psi for 15min- ok, connected control lines to both CIV, tested them with 4000psi- ok, observed CIV disks ruptured, worked on change rapture disks, tested again control line and CIVs with 4000psi for 15min- ok, continued run completion string as:

• 14 joints 3-1/2" tubing 10.2ppf, L-80, HSM-1

• XOS 4 1/2" HSM-1 x 3 1/2" HSM-1

• XOS 4-1/2" HSM-1 x TPG-2 p*b to 225m Pressure tested completion string with 8000psi for 15min- ok, rigged up slickline, ran 2.800" gauge ring to above check valve inside SSD, rigged down slick line equipment, continued run comp string with 4-1/2" tubing, 18.9ppf, L-80, TPG2 to 2146m, rigged up slick line, tried to set 2.750" check valve in 3-1/2" SSD- not succeed, pulled out to surface, set plug & prong in SSD, rigged up slickline, pressure tested completion string with 8000psi for 15min-ok, rigged up slick line, retrieved prong and plug from SSD, rigged slick line, continued run completion string with 4-1/2" tubing and:

• XOS TPG-2 pin * 4-1/2" HSM-1 box

• 4-1/2" pup joint 18.9ppf, L-80, HSM-1

• 4-1/2" flow coupling 18.9ppf, L-80, super 13Cr, HSM-1

• 4-1/2" TRSSV 18.9ppf, L-80 super 13Cr, HSM-1 (profile : 3.437")

• 4-1/2" flow coupling 18.9ppf, L-80, super 13Cr, HSM-1

• 4-1/2" pup joint 18.9ppf, L-80, HSM-1

• 4-1/2" fluted swage adapter 18.9ppf, L-80, super 13Cr V. top * HSM-1

• 11* 4-1/2" 10k, MSP tubing hanger

• 4-1/2" Acme pin * 4-1/2" HSM-1 box

• 4-1/2" pup joint 18.9ppf, L-80 HSM0-1 to 3860m Rigged up slickline, set 2.750" check valve in 3-1/2" SSD, tested completion string with 8000psi-not ok, rigged up slick line, retrieved 2.750” check valve from SSD, set plug and prong in 3-1/2" SSD, tested completion string at 3860m with 8000psi-

ok, ran slick line to retrieved prong from SSD- not succeed, pulled out to surface, observed 2" SB pulling and slick line accessories left in hole, to recover fish ran 2" SB pulling, retrieved prong from SSD plug, ran 2.5" impression block and checked top of fish, tried to recover 2" SB pulling several times with no success, modified 2" SB pulling and ran in hole to top of fish, worked on same, recovered fish.

Ran 2.750" GS pulling and retrieved plug from 3-1/2" SSD, rigged down slick line, ran in hole with 4-1/2" landing joint N. Vam, connected 1/4" control line to tubing hanger and tested it with 8000psi- ok, connected 2no 3/8" CIL to tubing hanger and tested with 500psi- ok, flush and cleaned THS with water, ran completion string to 3870.5m landed 4-1/2" x 11" x 4-1/2" tubing hanger in THS, string up 1.5m as space out to 3869m, rigged up slick line, ran 2.750" shifting tools to open and close 3-1/2" SSD, after 2 times ran 2.562" plug to set in No-go landing nipple, ran and set prong in 3-1/2" No-go landing nipple, pressurized inside completion string to 300psi to ensure 3-1/2" expansion joint is full open, increased pressure gradually to 3500psi and set 7" hydro permanent production packer, to test completion string increased pressure to 8000psi and held for 15min- ok and pull tested with 15klbs- ok.

Rigged up r/u slick line, tried to retrieve prong- No success, rigged down slick line, rigged up coiled tubing, flushed coil tubing and ran in hole with 1-1/2" coiled tubing to No-Go at 3860m, circulated and cleaned the completion string, pulled out to surface and rigged down coiled tubing equipment. Rigged up slick line, tried to retrieve prong- no success, rigged down slick line, rigged up coiled tubing, flushed coiled tubing and ran in hole with 1-1/2" coiled tubing to top of No-go at 3860m, circulated and cleaned completion string, pulled out to surface and laid down coiled tubing equipment, rigged up slick line, ran with SB pulling and retrieved plug & prong in 2 stage, ran with GS pulling and retrieved protection sleeve from TRSSV, closed 4-1/2" TRSSV, rigged down slick line, back off 4-1/2" landing joint, installed 4" 2-way valve in tubing hanger then closed blind rams and tested two way with 8000psi- ok, nipple down 13-5/8" 10k BOP stack, checked 11"x4-1/2" 10k hanger neck, control line and CIL- ok, tested body and valves of 10k x-mass tree before installation with 8000psi-ok, nipple up 11"x4-1/16" 10k adaptor flange with MSP BMV and tested s-seals with 7500psi- ok, tested control line with 5000psi- ok, nipple up MSP 4-1/16"*4-1/16"*4-1/16"*4-1/16" 10k x-mass tree, body tested xmass tree and MSP 11"*4-1/16" 10k BMV with 1000psi for 15min- ok, continued nipple up 11"x4-1/16" 10k adaptor flange with BMV, tested s-seals with 7500psi- ok,

tested control line with 5000psi- ok, nipple up and body tested x-mass tree and 11"x4-1/16" 10k MSP BMV with 1000psi for 15min- ok, tested metal seal of tubing hanger with 7500psi for 15min- ok, bleed off pressure to 500psi and closed 4-1/16" adaptor flanged, inflow tested- ok, pressure tested p-seals from section B with 1000psi- ok, tested p-seal from section C upper test port with 2500psi- ok, pressure tested lower test port -not ok, observed oil return from 9-5/8"x13-5/8" annulus, retrieved 4" 2-way valve from tubing hanger, set 4"one way valve in tubing hanger, closed 4-1/16" 5k TMV, 4-1/16" BMV, SSV, production and kill valves, released the rig at 2400 on 96/12/23.

Rig less operation was performed on date 17.05.1398.

Tried to RIH with coiled tubing to unload the well. There was an obstruction in the well at depth 3918m. Unable to pass the depth. Several runs were performed by coiled tubing and mill to remove the obstruction.

Rig less operation for unloading and well flowing was performed on date 30.01.1400.

Pressure monitoring was performed on tubing head spool vent port, it was 4,000 psi. Metal seal of tubing hanger is damaged.

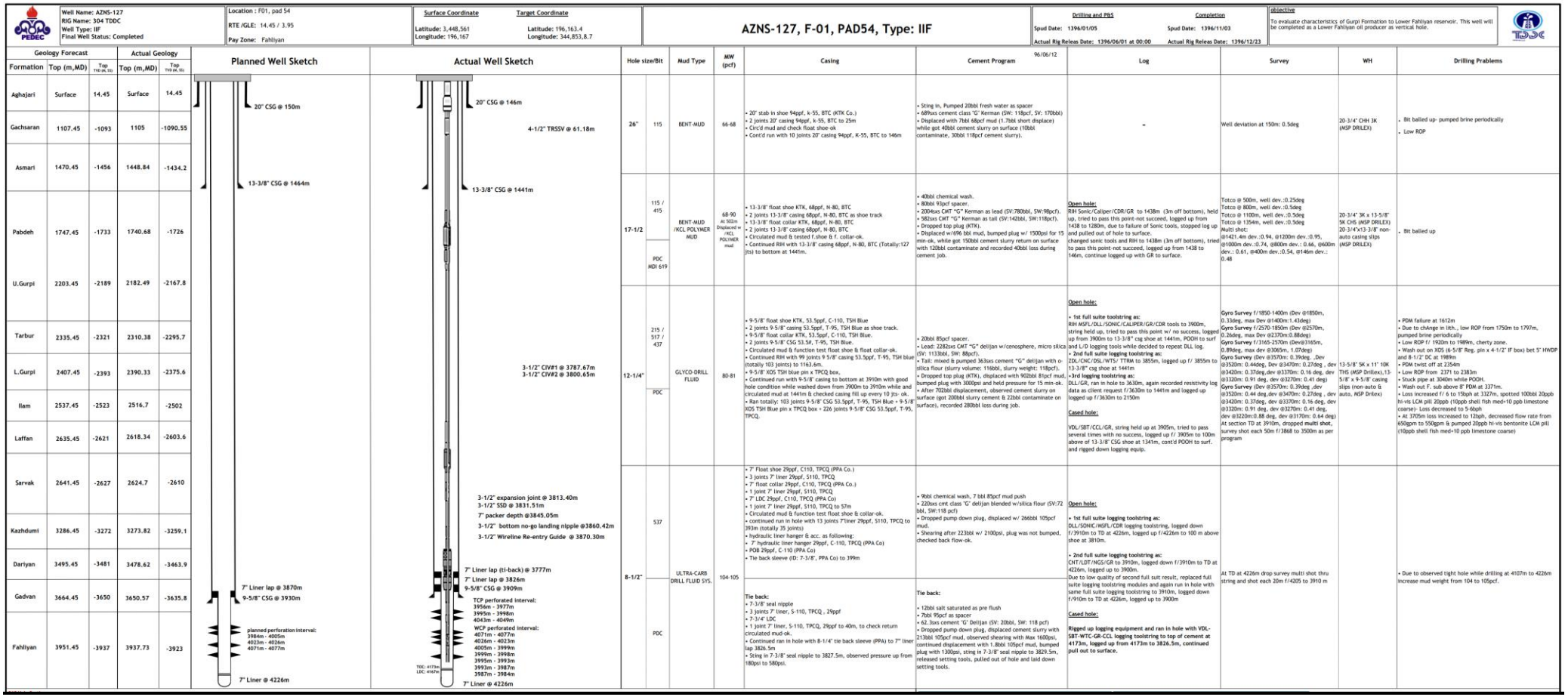
Try to run coiled tubing to TD and wash the wellbore with xylene. Run CCL/GR/MFC logging BHA in the well. It was observed 7” liner ID has been decreased to about 3inches in interval 3889m-3911m.

Flow the well, an abnormal annulus pressure was observed.

The other Rig less monitoring well annulus pressure on date 18.09.1400.

It was annulus pressure is 5,300 psi. Sampling was performed from annulus fluid. It was contained with gas and hydrocarbon.

1.8 Well Montage



Well Name: AZNS-127
RIG Name: 304 TDDC
Well Type: IIF
Final Well Status: Completed

Location : F01, pad 54
RTE /GLE: 14.45 / 3.95
Pay Zone: Fahliyan

Surface Coordinate: Latitude: 3,448,561 Longitude: 196,167
Target Coordinate: Latitude: 196,163.4 Longitude: 344,853,8.7

AZNS-127, F-01, PAD54, Type: IIF

Drilling and P&S: Spud Date: 1396/01/05; Actual Rig Release Date: 1396/06/01 at 00:00
Completion: Spud Date: 1396/11/03; Actual Rig Release Date: 1396/12/23

objective: To evaluate characteristics of Gurpi Formation to Lower Fahliyan reservoir. This well will be completed as a Lower Fahliyan oil producer as vertical hole.

96/06/12

Geology Forecast Formation	Top (m,MD)	Top TVD (M, SS)	Actual Geology Top (m,MD)	Top TVD (M, SS)
Aghajari	Surface	14.45	Surface	14.45
Gachsaran	1107.45	-1093	1105	-1090.55
Asmari	1470.45	-1456	1448.84	-1434.2
Pabdeh	1747.45	-1733	1740.68	-1726
U.Gurpi	2203.45	-2189	2182.49	-2167.8
Tarbur	2335.45	-2321	2310.38	-2295.7
L.Gurpi	2407.45	-2393	2390.33	-2375.6
Ilam	2537.45	-2523	2516.7	-2502
Laffan	2635.45	-2621	2618.34	-2603.6
Sarvak	2641.45	-2627	2624.7	-2610
Kazhdumi	3286.45	-3272	3273.82	-3259.1
Dariyan	3495.45	-3481	3478.62	-3463.9
Gadvan	3664.45	-3650	3650.57	-3635.8
Fahliyan	3951.45	-3937	3937.73	-3923

Planned Well Sketch:
- 20" CSG @ 150m
- 13-3/8" CSG @ 1464m
- 7" Liner lap @ 3870m
- 9-5/8" CSG @ 3930m
- planned perforation interval: 3984m - 4005m, 4023m - 4026m, 4071m - 4077m
- 7" Liner @ 4226m

Actual Well Sketch:
- 20" CSG @ 146m
- 4-1/2" TRSSV @ 61.18m
- 13-3/8" CSG @ 1441m
- 3-1/2" CIVF1 @ 3787.67m
- 3-1/2" CIVF2 @ 3800.65m
- 3-1/2" expansion joint @ 3813.40m
- 3-1/2" SSD @ 3831.51m
- 7" packer depth @3845.05m
- 3-1/2" bottom no-go landing nipple @3860.42m
- 3-1/2" Wireline Re-entry Guide @ 3870.30m
- 7" Liner lap (ti-back) @ 3777m
- 7" Liner lap @ 3826m
- 9-5/8" CSG @ 3909m
- TCP perforated interval: 3956m - 3977m, 3995m - 3998m, 4043m - 4049m
- WCP perforated interval: 4071m - 4077m, 4026m - 4023m, 4005m - 3999m, 3999m - 3998m, 3995m - 3993m, 3993m - 3987m, 3987m - 3984m
- TOC: 4173m, LOC: 4167m
- 7" Liner @ 4226m

Hole size/Bit		Mud Type	MW (pcf)	Casing	Cement Program	Log	Survey	WH	Drilling Problems
26"	115	BENT-MUD	66-68	• 20" stab in shoe 94ppf, k-55, BTC (KTK Co.) • 2 joints 20" casing 94ppf, k-55, BTC to 25m • Circ'd mud and check float shoe-ok • Cont'd run with 10 joints 20" casing 94ppf, K-55, BTC to 146m	• Sting in, Pumped 20bbl fresh water as spacer • 689sxs cement class 'G' Kerman (SW: 118pcf, SV: 170bbl) • Displaced with 7bbl 68pcf mud (1.7bbl short displace) while got 40bbl cement slurry on surface (10bbl contaminate, 30bbl 118pcf cement slurry).	-	Well deviation at 150m: 0.5deg	20-3/4" CHH 3K (MSP DRILEX)	• Bit balled up- pumped brine periodically • Low ROP
17-1/2	115 / 415 PDC MDI 619	BENT-MUD /KCL POLYMER MUD	68-90 At 502m Displaced w /KCL POLYMER mud	• 13-3/8" float shoe KTK, 68ppf, N-80, BTC • 2 joints 13-3/8" casing 68ppf, N-80, BTC shoe track • 13-3/8" float collar KTK, 68ppf, N-80, BTC • 2 joints 13-3/8" casing 68ppf, N-80, BTC • Circulated mud & tested f.shoe & f. collar-ok. • Continued RIH with 13-3/8" casing 68ppf, N-80, BTC (Totally:127 jts) to bottom at 1441m.	• 40bbl chemical wash. • 80bbl 93pcf spacer. • 2004sxs CMT "G" Kerman as lead (SV:780bbl, SW:98pcf). • 582sxs CMT "G" Kerman as tail (SV:142bbl, SW:118pcf). • Dropped top plug (KTK). • Displaced w/696 bbl mud, bumped plug w/ 1500psi for 15 min-ok, while got 150bbl cement slurry return on surface with 120bbl contaminate and recorded 40bbl loss during cement job.	Open hole: RIH Sonic/Caliper/CDR/GR to 1438m (3m off bottom), held up, tried to pass this point-not succeed, logged up from 1438 to 1280m, due to failure of Sonic tools, stopped log up and pulled out of hole to surface. changed sonic tools and RIH to 1438m (3m off bottom), tried to pass this point-not succeed, logged up from 1438 to 146m, continue logged up with GR to surface.	Totco @ 500m, well dev.:0.25deg Totco @ 800m, well dev.:0.5deg Totco @ 1100m, well dev.:0.5deg Totco @ 1354m, well dev.:0.5deg Multi shot: @1421.4m dev.:0.94, @1200m dev.:0.95, @1000m dev.:0.74, @800m dev.: 0.66, @600m dev.: 0.61, @400m dev.:0.54, @146m dev.: 0.48	20-3/4" 3K x 13-5/8" 5K CHS (MSP DRILEX) 20-3/4"x13-3/8" non-auto casing slips (MSP DRILEX)	• Bit balled up
12-1/4"	215 / 517 / 437 PDC	GLYCO-DRILL FLUID	80-81	• 9-5/8" float shoe KTK, 53.5ppf, C-110, TSH Blue • 2 joints 9-5/8" casing 53.5ppf, T-95, TSH Blue as shoe track. • 9-5/8" float collar KTK, 53.5ppf, C-110, TSH Blue. • 2 joints 9-5/8" CSG 53.5#, T-95, TSH Blue. • Circulated mud & function test float shoe & float collar-ok. • Continued RIH with 99 joints 9-5/8" casing 53.5ppf, T-95, TSH blue (totally 103 joints) to 1163.6m. • 9-5/8" XO5 TSH blue pin x TPCQ box, • Continued run with 9-5/8" casing to bottom at 3910m with good hole condition while washed down from 3900m to 3910m while and circulated mud at 1441m & checked casing fill up every 10 jts- ok. • Ran totally: 103 joints 9-5/8" CSG 53.5ppf, T-95, TSH Blue + 9-5/8 XO5 TSH Blue pin x TPCQ box + 226 joints 9-5/8" CSG 53.5ppf, T-95, TPCQ.	• 20bbl 85pcf spacer. • Lead: 2282sxs CMT "G" deli/jan w/cenosphere, micro silica (SV: 1133bbl, SW: 88pcf). • Tail: mixed & pumped 363sxs cement "G" deli/jan with o-silica flour (slurry volume: 116bbl, slurry weight: 118pcf). • Dropped top plug (KTK), displaced with 903bbl 81pcf mud, bumped plug with 3000psi and held pressure for 15 min-ok. • After 702bbl displacement, observed cement slurry on surface (got 208bbl slurry cement & 22bbl contaminate on surface), recorded 280bbl loss during job.	Open hole: • 1st full suite toolstring as: RIH MGFL/DLL/SONIC/CALIPER/GR/CDR tools to 3900m, string held up, tried to pass this point w/ no success, logged up from 3900m to 13-3/8" csg shoe at 1441m, POOH to surf and L/D logging tools while decided to repeat DLL log. • 2nd full suite logging toolstring as: ZDL/CNC/DSL/WTS/ TTRM to 3855m, logged up f/ 3855m to 13-3/8" csg shoe at 1441m • 3rd logging toolstring as: DLL/GR, ran in hole to 3630m, again recorded resistivity log data as client request f/3630m to 1441m and logged up logged up f/3630m to 2150m Cased hole: VDL/SBT/CCL/GR, string held up at 3905m, tried to pass several times with no success, logged up f/ 3905m to 100m above of 13-3/8" CSG shoe at 1341m, cont'd POOH to surf. and rigged down logging equip.	Gyro Survey f/1850-1400m (Dev @1850m, 0.33deg, max Dev @1400m:1.43deg) Gyro Survey f/2570-1850m (Dev @2570m, 0.26deg, max Dev @2370m:0.88deg) Gyro Survey f/3165-2570m (Dev@3165m, 0.89deg, max dev @3065m, 1.07deg) Gyro Survey (Dev @3570m: 0.39deg ,,Dev @3520m: 0.44deg, Dev @3470m: 0.27deg , dev @3420m: 0.37deg,dev @3370m: 0.16 deg, dev @3320m: 0.91 deg, dev @3270m: 0.41 deg) Gyro Survey (Dev @3570m: 0.39deg ,dev @3520m: 0.44 deg,dev @3470m: 0.27deg , dev @3420m: 0.37deg, dev @3370m: 0.16 deg, dev @3320m: 0.91 deg, dev @3270m: 0.41 deg, dev @3220m:0.88 deg, dev @3170m: 0.64 deg) At section TD at 3910m, dropped multi shot, survey shot each 50m f/3868 to 3500m as per program	13-5/8" 5K x 11" 10K THS (MSP Drillex),13-5/8" x 9-5/8" casing slips (non-auto & auto, MSP Drilex)	• PDM failure at 1612m • Due to change in lith., low ROP from 1750m to 1797m, pumped brine periodically • Low ROP f/ 1920m to 1989m, cherty zone. • Wash out on XO5 (6-5/8" Reg. pin x 4-1/2" IF box) bet 5' HWDP and 8-1/2" DC at 1989m • PDM twist off at 2354m • Low ROP from 2371 to 2383m • Stuck pipe at 3040m while POOH. • Wash out F. sub above 8" PDM at 3371m. • Loss increased f/ 6 to 15bph at 3327m, spotted 100bbl 20ppb hi-vis LCM pill 20ppb (10ppb shell fish med+10 ppb limestone coarse)- Loss decreased to 5-6bph • At 3705m loss increased to 12bph, decreased flow rate from 650gpm to 550gpm & pumped 20ppb hi-vis bentonite LCM pill (10ppb shell fish med+10 ppb limestone coarse)
8-1/2"	537 PDC	ULTRA-CARB DRILL FLUID SYS.	104-105	• 7" Float shoe 29ppf, C110, TPCQ (PPA Co.) • 3 joints 7" liner 29ppf, S110, TPCQ • 7" float collar 29ppf, C110, TPCQ (PPA Co.) • 1 joint 7" liner 29ppf, S110, TPCQ • 7" LDC 29ppf, C110, TPCQ (PPA Co) • 1 joint 7" liner 29ppf, S110, TPCQ to 57m • Circulated mud & function test float shoe & collar-ok. • continued run in hole with 13 joints 7"liner 29ppf, S110, TPCQ to 393m (totally 35 joints) • hydraulic liner hanger & acc. as following: • 7" hydraulic liner hanger 29ppf, C-110, TPCQ (PPA Co) • POB 29ppf, C-110 (PPA Co) • Tie back sleeve (ID: 7-3/8", PPA Co) to 399m Tie back: • 7-3/8" seal nipple • 3 joints 7" liner, S-110, TPCQ , 29ppf • 7-3/4" LDC • 1 joint 7" liner, S-110, TPCQ, 29ppf to 40m, to check return circulated mud-ok. • Continued ran in hole with 8-1/4" tie back sleeve (PPA) to 7" liner lap 3826.5m • Sting in 7-3/8" seal nipple to 3827.5m, observed pressure up from 180psi to 580psi.	• 9bbl chemical wash, 7 bbl 85pcf mud push • 220sxs cmt class 'G' deli/jan blended w/silica flour (SV:72 bbl, SW:118 pcf) • Dropped pump down plug, displaced w/ 266bbl 105pcf mud. • Shearing after 223bbl w/ 2100psi, plug was not bumped, checked back flow-ok. Tie back: • 12bbl salt saturated as pre flush • 7bbl 95pcf as spacer • 62.3sxs cement 'G' Deli/jan (SV: 20bbl, SW: 118 pcf) • Dropped pump down plug, displaced cement slurry with 213bbl 105pcf mud, observed shearing with Max 1600psi, continued displacement with 1.8bbl 105pcf mud, bumped plug with 1300psi, sting in 7-3/8" seal nipple to 3829.5m, released setting tools, pulled out of hole and laid down setting tools.	Open hole: • 1st full suite logging toolstring as: DLL/SONIC/MGFL/CDR logging toolstring, logged down f/3910m to TD at 4226m, logged up f/4226m to 100 m above shoe at 3810m. • 2nd full suite logging toolstring as: CNT/LDT/NGS/GR to 3910m, logged down f/3910m to TD at 4226m, logged up to 3900m. Due to low quality of second full suit result, replaced full suite logging toolstring modules and again run in hole with same full suite logging toolstring to 3910m, logged down f/910m to TD at 4226m, logged up to 3900m Cased hole: Rigged up logging equipment and ran in hole with VDL-SBT-WTC-GR-CCL logging toolstring to top of cement at 4173m, logged up from 4173m to 3826.5m, continued pull out to surface.	At TD at 4226m drop survey multi shot thru string and shot each 20m f/4205 to 3910 m		• Due to observed tight hole while drilling at 4107m to 4226m increase mud weight from 104 to 105pcf.

1.9 Casing & Liner Depth

Casing Spec.	20"	K-55, 94#, BTC	0-146 m
	13-3/8"	N-80, 68#, BTC	0-1441 m
	9-5/8"	103 joint 53.5ppf, T-95, TSH Blue, 226 joints 53.5ppf, T-95, TPCQ.	0-3909 m
	7"	110 S, 29#, TPCQ	3777 – 4226 m

Table 7: Casing & Liner Data

1.10 Well Schematic before Workover

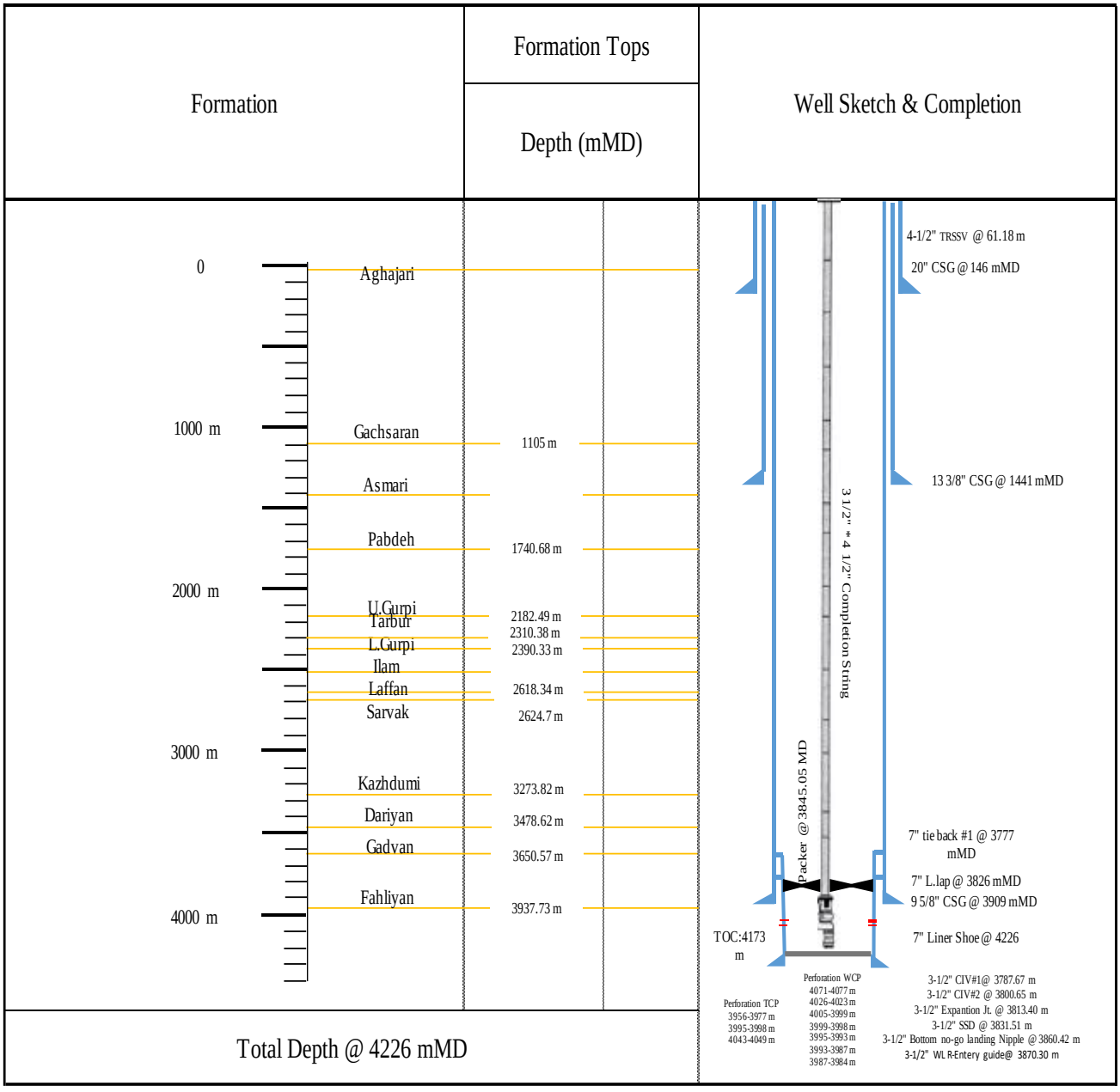


Figure 4: Well Schematic Before workover

1.11 Well Schematic after Workover

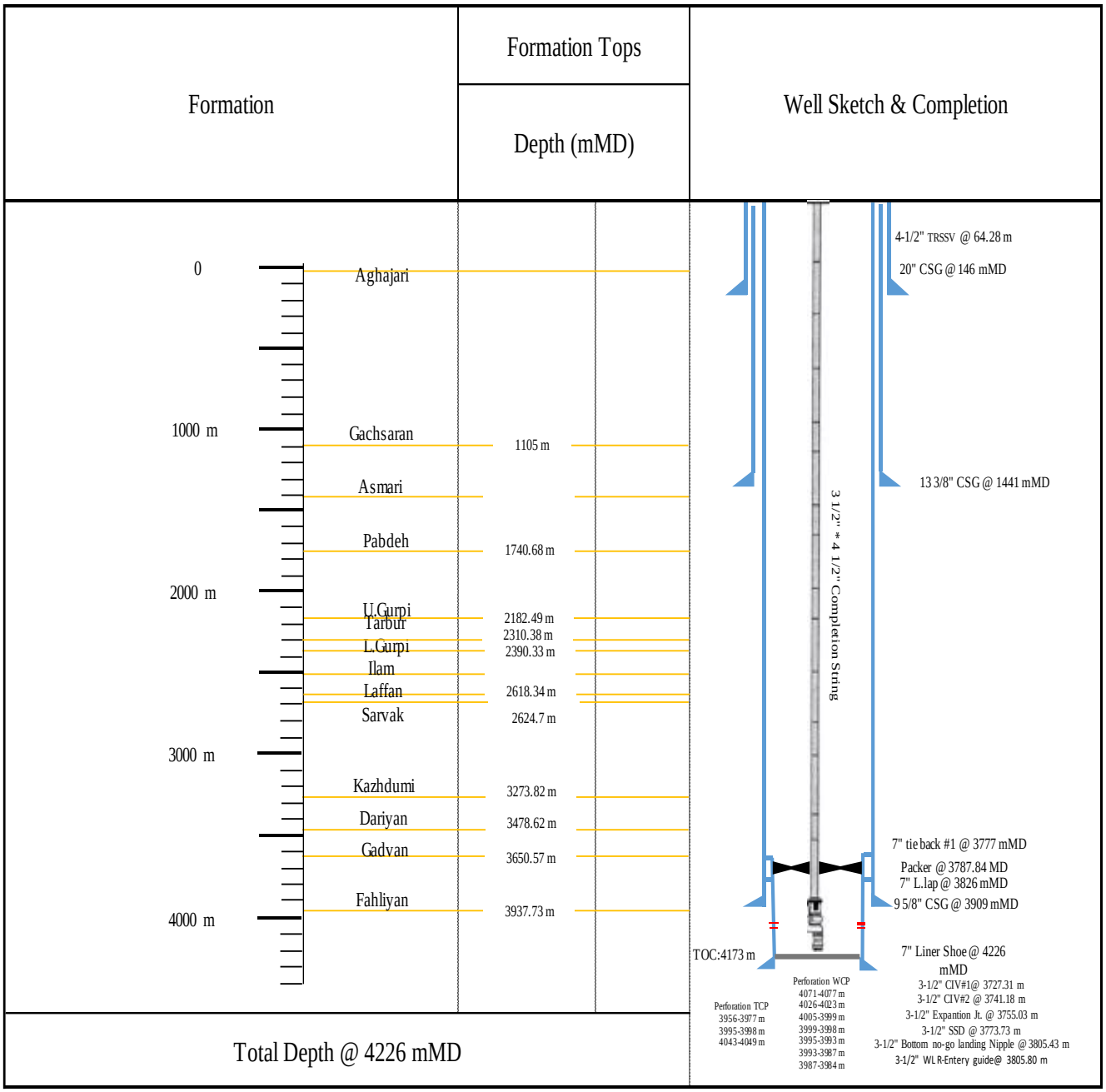


Figure 5: Well Schematic After workover

2. Geological Information

2.1 Formation Stratigraphy and Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
Miocene	Fars	Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi / Tarbur / L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
	Bangestan	Laffan		Claystone	2691	2642	2625
Turonian Cenomanian	Bangestan	Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian	Bangestan	Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian	Khami	Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian	Khami	Fahliyan		Limestone, Marl	4092	3989	3974

Figure 6: Azadegan Stratigraphic Column

3. Logging Program

Hole Size	Casing Size	Formation	Logs
			Cased Hole
12-1/4"	9-5/8" CSG.	Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan	Corrosion Log GR-CCL (perforated interval)
8-1/2"	7"L.	Fahliyan	

Table 8: Logging Program

4. Program Outline

4.1 Contractors Involved

Table 9: Contractor Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC 208
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging and Perforation	SEALAND
5	Mud Logging	MAPSA KISH
6	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
7	Casing and Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST and RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC

4.2 Time Breakdown

De-Complete, Re-Complete Operation TBD of Well: AZNS-127 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over									
Held PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC.	Trip	4226	48	2.00	2.0	2.2	52.8	2.20	2.20
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.	Wellhead	4226	48	2.00	4.0	2.2	52.8	2.20	4.4
Secure the well by close 3SV and X-Mass Tree Valves, Remove 7-1/16" XMT cap, R/U Slick line and perform pressure test,RIH w/Gauge Cutter to top of 3SV, RIH & set protection sleeve, RIH w/Gauge Cutter to guide shoe.Note: If unable to pass to the bottom, follow contingency plan #1. Note:Chemical Wash May be Needed while RIH W/CT.	Trip	4226	40	1.67	5.7	1.8	44.0	1.83	6.2
try to displace well with heavy mud weight through SSD, and/or R/U CT and RIH to btm.displace the well with heavy weitgh mud .R/U Slick Line, Retrieve Protection Sleeve, Close 3SV, R/D S/L.	Trip	4226	50	2.08	7.8	2.3	55.0	2.29	8.5
Set TWCV inside tubing hanger & pressure test, N/D XMT & N/U BOPs, open tie down screws,perform BOP pressure Test Note: if observe any leakage or damage on THS, refer to next sequence	Wellhead	4226	60	2.50	10.3	2.8	66.0	2.75	11.3
Retrieve TWCV, R/U casing running equipment, RIH with TBH retrieving tool & engage with TBH, apply over pull/Rotation to un-latch the seal assembly. circulate to balance mud weight and flow check before pulling out. Note: if observe any damage on THS, follow contingency plan#2 Note: If unable to un-latch seal assembly, follow contingency plan #3.	Completion	4226	50	2.08	12.3	2.3	55.0	2.29	13.6
Pull out the completion string from anchor seal and lay down completion string precautionary due to probable trapped pressure below plug & Prong. While POOH monitor the well and annulus to make sure the well is safe and over balance.	Trip	4226	90	3.75	16.1	4.1	99.0	4.13	17.7
Perform BOP Test.	BOP	4226	6	0.25	16.3	0.3	6.6	0.28	18.0
RIH with 6" mill assembly & mill permanent packer, chase to bottom of 7" liner (or where ever packer stopped), POOH.	Trip	4226	216	9.00	25.3	9.9	237.6	9.90	27.9
RIH with taper tap & catch the packer, recover the completion packer, POOH, L/D BHA.	Trip	4226	70	2.92	28.3	3.2	77.0	3.21	31.1
RIH with 8-1/2" bit and clean out BHA to 7" liner lap@ 3777m.CBU. POOH to surface.	Log	4226	48	2.00	30.3	2.2	52.8	2.20	33.3
Perform BOP Test.	BOP	4226	6	0.25	30.5	0.3	6.6	0.28	33.6
RIH with 5-7/8" to 6" mill and clean out BHA to TOC@ 4170m.CBU till hole clean. POOH to surface. Note: If no success to pass interval 3889m-3911m., follow contingency plan #4.	Log	4226	175	7.29	37.8	8.0	192.5	8.02	41.6
Upon client approval RIH with 6" Mill and BHA, milling & dress pre-perforation interval.CBU till hole clean. POOH to surface.	Log	4226	0	0.00	37.8	0.0	0.0	0.00	41.6
RIH with 9-5/8" * 7" tandem scraper BHA, CBU & POOH to surface.	Trip	4226	36	1.50	39.3	1.7	39.6	1.65	43.2
RIH with wireline corrosion logging tools to bottom, log up to surface in two separate run (7" Liner & 9-5/8" Casing).	log	4226	40	1.67	40.958333	1.83	44.0	1.8	45.1
Perform BOP Test.	BOP	4226	6	0.25	41.2	0.3	6.6	0.28	45.3
RIH with GR-CCL log to confirm perforation intervals	log	4226	24	1.00	42.2	1.10	26.4	1.1	46.4
RIH with 6" mill & clean out BHA (7" scraper included) to bottom , displace the well with completion fluid (non-solid), POOH & L/D DP's.	Trip	4226	72.00	3.0	45.2	3.3	79.20	3.3	49.7
Run 10K completion string as per program. NOTE: Annulus pressure test during completion operation is related to corrosion log result.	Completion	4226	125	5.21	50.4	5.7	137.5	5.73	55.5
Set BPV inside TBH, N/D BOP's & install new 10K XMT.	Completion	4226	48	2.00	52.4	2.2	52.8	2.20	57.7
Retrive BPV, install TWCV and Pressure test the XMT.	Completion	4226	24	1.00	53.4	1.1	26.4	1.10	58.8
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.	Completion	4226	10	0.42	53.8	0.5	11.0	0.46	59.2

Retrieve TWCV,R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
RIH with CT to bottom, displace the well with diesel or nitrogen & observ well, POOH.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs. Note 1: The time is only for flowing duration (R/U, R/D & preparation time is excluded). Note 2: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
RIH with S/L, set protection Sleeve, POOH & R/D SL.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
RIH with C/T to bottom & displace the well with diesel, POOH.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
RIH with S/L, install plug & prong into NO GO landing nipple, POOH.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
RIH with S/L, retrieve protection sleeve.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion	4226	Rig Less	0.00	53.8	0.0	0.0	0.00	59.2
Total Duration of Work-over:				***53.8***	***53.8***			***59.2***	***59.2***
NOTE:									
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip	4226	10	0.42	54.3	0.5	11.0	0.46	59.7
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service	4226	15.00	0.63	54.9	0.7	16.5	0.69	60.4
Total Duration of AZNS-127 Work-over Well:					54.9				60.4

Contingency Plans TBD of Well: AZNS-127 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%)	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
						10			
Based on operation situation in each step, the following contingency plans might be followed									
Contingency plan #1: if observe any obstruction while each run w/ Gauge Cutter RIH w/CT and wash from obstruction depth to 3SV/No-GO Landing Nipple with Zylyne.	Trip	4226	96	4.00	4.0	4.4	105.6	4.40	4.4
Contingency plan #2: if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG). Note:Proper Equipment shall be provided by client and install by contractor.	Completion	4226	200.0	8.3	12.3	9.2	220.0	9.2	13.6
Contingency plan #3: RIH w/ colliding tools & collide tubing f/ top of packer, then mill tubing & packer.	Completion	4226	320.0	13.3	25.7	14.7	352.0	14.7	28.2
Contingency plan #4: RIH w/ propper tapper mill and work on interval to smooth obstraction.	Completion	4226	500.0	20.8	46.5	22.9	550.0	22.9	51.2
Contingency plan #5: as per client request perform dry test and reated to result run and set SN. Packer.	Completion	4226	170.0	7.1	53.6	7.8	187.0	7.8	58.9
Total Duration of AZNS-127 Work-over Well:					53.6				58.9

5. Sequence of Operation

5.1 Main Objective

Decision has been made to de-complete he well & re-complete with new completion equipment. Check and smooth collapse interval from 3889-3911m.

5.2 Main plan

1. Held PJSM. Spud the Rig. Stand Back required 5" & 3-1/2" DPs, 5" HWDPs and 6-1/2" DC .
2. Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.
3. Secure the well by close 3SV and X-Mass Tree Valves, Remove 7-1/16"" XMT cap, R/U Slick line and perform pressure test, RIH w/Gauge Cutter to top of 3SV, RIH & set protection sleeve, RIH w/Gauge Cutter to guide shoe.

   Note: If unable to pass to the bottom, follow contingency plan #1.

   Note: Chemical Wash May be needed while RIH W/CT".
4. Try to displace well with heavy mud weight through SSD, and/or R/U CT and RIH to BTM. Displace the well with heavy weitgh mud .R/U Slick Line, Retrieve Protection Sleeve, Close 3SV, R/D S/L .
5. Set TWCV inside tubing hanger & pressure test, N/D XMT & N/U BOPs, open tie down screws, perform BOP pressure Test

   Note: if observe any leakage or damage on THS, refer to next sequence.
6. Retrieve TWCV, R/U casing running equipment, RIH with TBH retrieving tool & engage with TBH, apply over pull/Rotation to un-latch the seal assembly. Circulate to balance mud weight and flow check before pulling out.

   Note: if observe any damage on THS, follow contingency plan#2

   Note: If unable to un-latch seal assembly, follow contingency plan #3

7. Pull out the completion string from anchor seal and lay down completion string precautionary due to probable trapped pressure below plug & Prong. While POOH monitor the well and annulus to make sure the well is safe and over balance.
8. Perform BOP Test.
9. RIH with 6" mill assembly & mill permanent packer, chase to bottom of 7" liner (or where ever packer stopped), POOH.
10. RIH with taper tap & catch the packer, recover the completion packer, POOH, L/D BHA.
11. RIH with 8-1/2" bit and clean out BHA to 7" liner lap@ 3777m.CBU. POOH to surface.
12. Perform BOP Test.
13. RIH with 5-7/8" to 6" mill and clean out BHA to TOC@ 4170m.CBU till hole clean. POOH to surface.

    Note: If no success to pass interval 3889m-3911m. Follow contingency plan #4.

    Note: In case of any damage to existing casing or side track while milling the collapsed interval remedial action should be approved by client and will be done by contractor as extra work.
14. Upon client approval RIH with 6" Mill and BHA, milling & dress pre-perforation interval. CBU till hole clean. POOH to surface.
15. RIH with 9-5/8" * 7" tandem scraper BHA, CBU & POOH to surface.
16. RIH with wireline corrosion logging tools to bottom, log up to surface in two separate run (7" Liner & 9-5/8" Casing).
17. Perform BOP Test.
18. RIH with GR-CCL log to confirm perforation intervals
19. RIH with 6" mill & clean out BHA (7" scraper included) to bottom, displace the well with completion fluid (non-solid), POOH & L/D DP's.
20. Run 10K completion string as per program.

NOTE: Annulus pressure test during completion operation is related to corrosion log result.

21. Set BPV inside TBH, N/D BOP's & install new 10K XMT.

22. Retrieve BPV, install TWCV and Pressure test the XMT.
23. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.
24. Retrieve TWCV, R/U Slick Line, Set protection sleeve, retrieve plug & prong from NO GO landing nipple, POOH.
25. RIH with CT to bottom, displace the well with diesel or nitrogen & observe well, POOH.
26. R/U SL, RIH & retrieve protection sleeve, POOH, R/D SL.
27. Flow the well as pre-acid clean up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.
28. R/U & RIH S.L, Set protection sleeve, POOH & R/D S.L
29. RIH with C/T & perform acid wash as per program, POOH to surface. RIH w/SL and retrieve protection sleeve"
30. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for min 96 hrs.

    Note: The time is only for flowing duration (R/U, R/D & preparation time is excluded).

    Note: Acidizing & post-acid clean-up maybe perform as rig less job upon contractor request & client approval but in case of any failure in well integrity the contractor should provide drilling rig & required services in own responsibility & fix the problem.
31. RIH with S/L, set protection Sleeve, POOH & R/D SL.
32. RIH with C/T to bottom & displace the well with diesel, POOH.
33. RIH with S/L, install plug & prong into NO GO landing nipple, POOH.
34. RIH with S/L, retrieve protection sleeve.
35. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

Contingency Plans of Well AZNS-127 Work-over:

Based on operation situation in each step, the following contingency plans might be followed:

Contingency plan #1: if observe any obstruction while each run w/ Gauge Cutter RIH w/CT and wash from obstruction depth to 3SV/No-GO Landing Nipple with Zylyne.

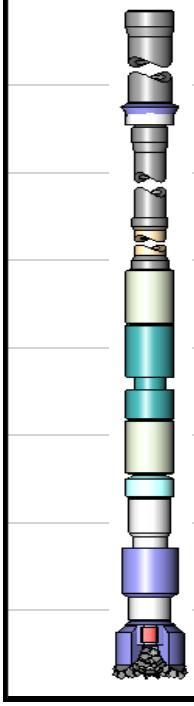
Contingency plan #2: if observe any leakage and or failure on THS, secure the well by barriers (RTTS / CEMENT PLUG).
Note: Proper Equipment shall be provided by client and install by contractor.

Contingency plan #3: RIH w/ colliding tools & collide tubing f/ top of packer, then mill tubing & packer.
Contingency plan #4: RIH w/ proper tapper mill and work on interval to smooth obstruction.
Contingency plan #5: as per client request perform dry test and reated to result run and set SN. Packer.

6. BHA Program

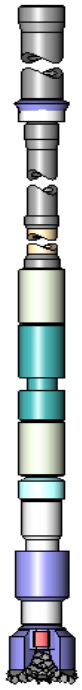
6.1 8-1/2" BHA

			Collar	OD
BIT SIZE (inch)	8-1/2"	BHA Configuration to C/Out Inside 9-5/8" Casing		
Depth In/Out (m)			DC-1	6-1/2"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	24,624			
Bouyed Weight (lb)	19,297			
Bouyed Weight Below Jar (lb)	14,267			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	3	5	3	4-1/2" IF Pin x Box	27.42	182.71
	XOS	1	6-1/2	3	4-1/2" Reg. P x 4-1/2" IF B	1	1
	6-1/2" Drill Collar	3	6-1/2	2-13/16	4-1/2" Reg. P x B	27.42	155.29
	6-1/2" Jar	1	6-1/2	2.75	4-1/2" Reg. P x B	8	127.87
	6-1/2" Drill Collar	13	6-1/2	2-13/16	4-1/2" Reg. P x B	118.82	119.87
	6-1/2" Bit Sub W/ Float Valve	1	6-3/4"	3	4-1/2" Reg. B x B	0.75	1.05
	8-1/2" MT Bit	1	8-1/2"	-	3-1/2" Reg. P x B	0.3	0.3

6.2 5-7/8" or 6" BHA

Mill SIZE (inch)	5-7/8" or 6"	**BHA Configuration to C/Out Inside 7" Liner**				**Collar**	**OD**
Depth In/Out (m)						DC-1	4-3/4"
HOLE INCLINATION	0						
MUD WEIGHT (pcf)	106						
B.FACTOR	0.784						
Weight in Air (lb)	29,642						
Bouyed Weight (lb)	23,229						
Bouyed Weight Below Jar (lb)	16,462						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	6" Mill	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

7. Mud & Cement Program

Note: Detailed mud & cement program, will be issued in due time as per client approved template.

7.1 Workover Hole Section

Workover section	7" Liner
From: 3777m	TO: 4226m
Kill the well, POOH of Completion String, Complete the hole with Completion 10K & Completion Fluid	

7.1.1 Overview

Drilling Fluid System	Kill Mud and Completion Fluid
Key products	Workover Fluid , Completion Fluid, Kill Mud
Water type	Drill Water
Solid Control	Shale Shaker
Potential problems	Down Hole losses , gas/oil influx, H2S,

Interval Mud Properties						
Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
0-4226	103-106	30-55	8-22	NC	9-10.5	45-55
Interval Completion Fluid Properties						

Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
Surface~3826	104	ALAP	N/A	NC	6-7.5	27-30

7.1.2 Mud Discussion & Mixing Procedure

7.1.2.1 Mud type

Kill mud 103-106 pcf will be used to kill the well and well will be displaced by Completion fluid of CaBr2 104pcf. Attention must be paid that there is Calcium Bromide Brine inside the well and all of them must be retrieved in separated mud tank by the time of displacing the well by kill mud. (As what the hole dictates, mud properties may be varied by the time).

7.1.2.2 Drilling Fluid Formulation

Kill Mud/Work over Mud formulation		
Component	**Conc. (ppb)**	**Component Description**
Drill water	0.5766 bbl/bbl	Base Fluid
Caustic Soda	0.3	PH Modifier (lime can be used instead)
Soda Ash	0.3	Hardness Remover and
Sodium Chloride	63	Non-Solid weighting Agent
Xanthan-Gum	0.2	Viscosifier (In case of need and based on situation can be used)
Calcium Carbonate	210	Weight Agent
Hematite	As needed	Weight Agent

*Above calculation is based on mud weight 103-106 pcf

*Lime can be substituted for caustic soda

Completion Fluid Formulation		
Component	**Conc. (ppb)**	**Component Description**
Calcium Bromide (14.2 ppg)	0.984bbl/bbl	Base Fluid
Corrosion Inhibitor	1% Vol	Anti-Corrosion
H2S Scavenger Liquid	0.5% Vol	H2S Removal/ Anti Corrosion
Oxygen Scavenger CA	0.5 ppb	Oxygen Removal/Anto Corrosion

The performance requirements designed for the completion fluid system are as follows:

- Recovered permeability: ≥ 90 ٪
- Corrosion rate ＜0.076 mm/year
- Diameter of solid particle ≤ 2μm
- For fluid pumping in the well NTU<30
- For fluid returning from the well: NTU <150
- Acid Soluble/ Removable with Solvent

7.1.2.3 Drilling Fluid Formulation

Kill Mud Mixing Procedure:

1. Fill tanks with calculated amount of Drill Water 0.576 bbl/bbl.
2. Treat Drill Water with Caustic soda and Soda Ash by 0.3pbb.
3. Mix NaCl at 66 ppb.
4. Add Xanthan-Gum by 0.2pbb.

5. Add CaCO3-F (210ppb) to reach MW of ±95 pcf.
6. Add Hematite (as needed) to reach MW of ±103 pcf.

   - In case of foaming use Alcohol-Based Defoamer.

Note: Use Hi Vis pill to clean hole if necessary.
Calculation is based on per bbl of mud.

Completion Fluid Mixing Procedure:

1. Make sure that all pits and lines involved with completion fluid are thoroughly cleaned, flushed and free of any fluid and solid.

   - Make sure that at least one large pit is kept empty for transferring ready brine into after given time for settling. After transferring one pit to empty pit, the first one is to be flushed and cleaned for accommodating the brine of next ready pit, and so on to the last pit.

   - Depends on the rig facilities and structure, mud engineer on board may find it better to prepare the fluids pit-by-pit.

2. Transfer the content of CaBr2 drums into designated pits.

   - Record the number of drums poured in each tank for accurate volume in it as the base of calculation for adding other additives into; do not rely on pit gauge, sounding or mud logging sensor, but have a look on them for comparison and/or correlation if found useful.

3. Mix Corrosion Inhibitor into pits at concentration of 3.5pbb (1% by volume).
4. Mix H2S Scavenger Liquid into pits at concentration of 0.5% by Volume.
5. Do not add Oxygen Scavenger CA at this stage as it should be added directly from the top of the pit, just prior to pump the fluid into the well.
6. Shut down all agitators and mixing pumps, and leave the mixed fluid in the pits for at least 12 hours to let any solid particle inside (mostly insoluble Lime) settles down.
7. Start transferring the clear fluid of the first pit to the empty pit by means of an air-powered Wilden pump.

   - Make sure that the suction hose of the Wilden pump is not put deep close to the bottom of the pit, otherwise, settled solids are sucked in.

8. After finishing the first pit, stop Wilden pump, wash, flush and clean the pit, use it as the empty pit for accommodating the fluid of second pit using the and same above method.

Note: all CaBr2 brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of CaBr2.

Based on well readings and the operation, mud weight may be varied.

7.1.2.4 Downhole Losses

As perforation had been done in well AZN#122 and there is exposure to reservoir, risk of any probable formation losses must be taken into consideration.
Losses will be minimized/avoided by the planned use of properly sized bridging/weight material (and Polymer) in conjunction with appropriate drilling practices. If seepage losses are noted on the well no specific action will be taken but close monitoring of the loss rate is to be made. For moderate loss rates addition of CaCO3 50 micron and 150 micron is recommended to ensure that there is adequate correctly sized material to act as bridging agents.

Seepage losses Less than 10 bbl/hr.

No specific action will be taken, but close monitoring of the probable loss rate is made.

Losses from 10-30 bbl/hr.

- Load CaCO3-M (150-300 micron) in system up to 5 ppb

***If loss not cured and even tends to increase, then:**

- Pump 40-60 bbl of pill made by 7-10 ppb CaCO3 150-300µ & 7-10 ppb CaCO3 600µ

Losses from 30-50 bbl/hr.

- Load CaCO3-M (150-300 micron) in system up to 5 ppb
- Since some of the CaCO3-M will be discarded over the shakers, CaCO3-M to compensate the separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60 bbl of pill made by
- 10-20 ppb CaCO3 200-300µ
- 10-20 ppb CaCO3 600 µ
- 10pbb Oyster Shell-F
- 10pbb Oyster Shell-M.

Losses from 50-100 bbl/hr.

- Load CaCO3-M (150-300 micron) in system up to 5 ppb
- Load CaCO3-C (600-100 micron) in system up to 5 ppb
- Since some of the CaCO3-M&C will be discarded over the shakers, add CaCO3-

M&C to compensate the separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60 bbl of the following pill:

- 10-20 ppb CaCO3 200-300 micron
- 10-20 ppb CaCO3 600 micron
- 7-10 ppb of each CaCO3 1000 micron & CaCO3 2000 micron
- 5-10 pbb Oyster Shell -F
- 5-10 pbb Oyster Shell -M

Losses more than 100 bbl/hr.

High concentrated LCM pill of various sizes of Calcium Carbonate and Oyster Shell-F&M&C should be pumped and spotted and gently squeezed over the loss zone.

Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30 minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close BOP and squeeze 10 bbl LCM pill into formation without exceeding 200 psi.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50 bbl of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20 bbl of LCM pill into the formation without exceeding 200 psi.
6. If a squeeze pressure develops, hold the pressure for 1 hour before running the drill string back to bottom.

Notes:

- LCM pills must be thoroughly homogenised and viscous enough before being pumped.
- Pumping high-concentrated LCM pill is subject to use of open-end DP or Mule-Shoe.
- Remove filter screens from the drill pipe (if there is any) and if necessary, remove strainers from the mud pump suction lines.

7.1.2.5 Engineering Comments

Keep watching and control on well to record loss.
All CaBr2 brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of CaBr2.

7.1.2.6 Drilling Issues

There is no drilling objective in AZN-122 workover well; mud should be compatible with reservoir and work over operation.
Solid Control equipment

- Use shale shaker to remove debris and LCM particle in case of use.
- Personal Safety

7.1.2.7 Mud Type

Inhibited CaBr2 brine (Calcium Bromide Brine) will be applied as completion fluid. And simple salt saturated kill mud to kill the well and clean out inside 9 5/8", 7" Liner.

Displacement Design
The combined effect of scrapers being run in the hole, the high flow rate will help to remove/disperse general debris.
At least 150 m of pill/spacer coverage or 3 min contact time at the highest pump rates should be considered.
Saturated CaCl2 brine can be applied as spacer/Hi-Vis pill while displacing.
Pit levels should be monitor throughout the displacement operation to catch and prevent an influx entering the well contributing to an incident.
All CaBr2 brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of CaBr2.

No.	Description	Volume [bbl.]	Comment
1	Viscous pill/spacer	50	Clean pill
2	Displace to inhibited CaBr2 Brine at 104pcf (Completion Fluid)		
3	Continue pumping Completion Fluid until cleanliness specification has been met.		

All CaBr2 retrieved from well must be stored in separated tanks. All CaBr2 left on surface, even dead volume of tanks, must be transferred back to drums at end of well. Not even a drop

of Cabr2 is not allowed to be wasted or dump in whole operation duration.
It is recommended to use desander tank as active pit of CaBr2 to minimize dead volume and return path.

7.1.2.8 Volume accounting & estimated required material

Work over /Kill Mud Fluid				AZADEGAN #122
Fluid Density (pcf)	103-106		Fluid System	Water base Work Over Fluid /Salt Saturated
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3909		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	4226
Liner Depth (m)	4226		Liner Lap (m)	3777
Liner OD / ID (inch)	7	6.276	Liner Length (m)	449

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	350
Cased Hole Volue [Capacity of 3782m 9⅝" Casing + Capacity of 426m 7" Liner]	960
Total New Volume Required (bbl)	**1310**

Estimated Chemical Consumption and Cost based on 103-106pcf MW							
PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Drill Water	bbl	N/A	0.576 bbl/bbl	755		755	N/A
Caustic Soda	25 Kg	N/A	0.3	7	25	32	N/A
Soda Ash	50 Kg	N/A	0.3	4	16	20	N/A
Sodium Chloride	1 MT	N/A	63	37	18	55	N/A
Xanthan-Gum	25 Kg	N/A	0.2	5	20	25	N/A
Calcium Carbonate	1 MT	N/A	325	194	30	224	N/A

*Above calculation are based on normal hole condition
*Above Calculated is Based on 103-**106pcf**

Completion Fluid			AZADEGAN #122	
Fluid Density (pcf)	104		Fluid System	CaBr2 Brine
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3909		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	4226
Liner Depth (m)	4226		Liner Lap (m)	3777
Liner OD / ID (inch)	7	6.276	Liner Length (m)	449

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	250
Cased Hole Volume [Capacity of 3782m 9⅝" Casing + Capacity of 426m 7" Liner]	960
Total New Volume Required (bbl)	**1210**

Estimated Chemical Consumption and Cost based on 74-75pcf MW							
PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Calcium Bromide (14.2 ppg)	55 Gal	N/A	0.984bbl/bbl	910	62	972	N/A
Corrosion Inhibitor	55 Gal	N/A	1% Vol	9	1	10	N/A
H2S Scavenger Liquid	55 Gal	N/A	0.5% Vol	5	3	8	N/A
Oxygen Scavenger CA	25 Kg	N/A	0.5 ppb	11	6	17	N/A

*Above calculation are based on normal hole condition

*Above Calculated is based on **104pcf**

7.1.2.9 Estimated contingency material

During all the phase keep a minimum safety stock of below items:

Chemical Type	Unit	Qty	Function
$CaCO_3$ M & C	1 MT BB	10 of each	Loss Circulation Material
Pipe-Freeing Agent	55 Gal DM	4	Environment-Friendly Stuck Pipe Agent
Lime	25 Kg Sx	80	CO_2 Contamination

Zinc Carbonate	25 Kg Sx	120	H_2S Scavenger
Sodium Bicarbonate	25 Kg Sx	40	Cement Contamination
Citric Acid	25 Kg Sx	40	Cement Contamination
Defoamer A	55 Gal DM	4	Anti-Foam
Bio-Cide	55 Gal DM	4	Anti-Bacteria
Oyster shell F/M/C	25 Kg Sxs	160each	Loss circulation material
Drilling Detergent	55Gal	4	Bit Balling Reducer
Oxygen Scavenger S	25Kg	30	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	55Gal	4	Polymer Temperature Stabilizer
Corrosion Inhibitor	55Gal	4	Corrosion Inhibitor
Lubricant	55Gal	8	Biodegradable lubricator

7.1.2.10 Material requirements for each well

Item	QTY	Unit	Description	Function
1	972	55 Gal Dr	Calcium Bromide (14.2 ppg)	CaBr2 brine
2	55	1 MT	Salt	Weighting Agent
3	224	1 MT	Calcium Carbonate 75 Mic	Weighting Agent
4	32	25 Kg Sx	Caustic Soda	PH Control
5	20	50 Kg Sx	Soda Ash	Hardness reducer
6	25	25 Kg Sx	Xanthan-Gum	Viscosifier
9	80	25 Kg Sx	Lime	PH Control
10	160	25 Kg Sx	Shell Fish	LCM
11	160	25 KG Sx	Limestone Chips	LCM
12	10	55 Gal Dr	Corrosion Inhibitor	Completion Fluid Corrosion Inhibitor
13	8	55 Gal Dr	H2S SCAV	Completion Fluid H2S SCAV

14	17	25 Kg Sx	OX SCAV	Completion Fluid OX SCAV
15	1	55 Gal Dr	BIO-CIDE	Anti-bacterial
16	12	55 Gal Dr	PH BUFFER	PH Stabilizer

7.2 Cement Program

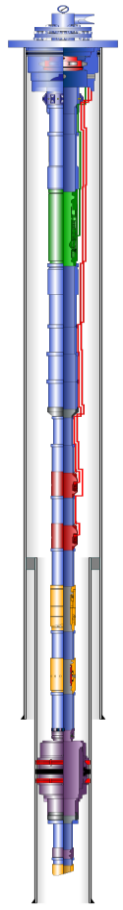
Not applicable.

8. Completion

8.1 Proposed Completion string

AZNS-127 Completion Sketch

Contractor		Liner & Tubing Data					Client	
Field	AZADEGAN	Description	7"Liner	3 1/2" Tubing	4 1/2" Tubing	Tubing	7" Tie back Liner Lap:	3777
Well Number:	AZNS-127	OD[in]	7.00	3.50	4.50		7" Liner lap:	3826
Rig :	OEOC-208	Nominal weight [lb/ft]	29.00	10.20	18.90		5' Liner Shoe:	
RTE-GLE	13.45	Grade	C-110	L-80	L-80		5" Liner Shoe	-
RTE-THF	3.95	Thread	TPCQ	HSM3	HSM3		Total Depth:	4226
Rig Spudded Date :	TBA	Inside diameter [in]	6.184	2.92	3.640		X-mass Tree:	MSP-T TYPE
End of Operation	TBA	API Drift Diameter	6.059	2.80	3.515		Size & Rating:	4 1/16"-10K



Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)				11.69	0.000
	2	TBG HANGER 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	3.68	0.6	12.29	11.690
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI	6.1	3.64	1.01	13.300	12.290
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.64	60.940	13.300
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	62.560	60.940
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT	5.5	3.64	1.72	64.280	62.560
	7	TRSSV-10 Tubing Retrievable Saftey Valve, 4 1/2", 18.9# W/ 3.437" Seal bore,HSM3 , SUPER 13CR-110KSI, 10K WP With AF Profile, Flapper Type,	7.5	3.437	1.78	66.060	64.280
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT	5.5	3.64	1.11	67.170	66.060
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	68.790	67.170
	10	381 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	3627.12	3695.910	68.790
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1.02	3696.930	3695.910
	12	3 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	28.56	3725.490	3696.930
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3727.310	3725.490
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3728.040	3727.310
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3729.860	3728.040
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3739.360	3729.860
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3741.180	3739.360
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	3741.910	3741.180
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.8	3743.710	3741.910
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3753.210	3743.710
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	3755.030	3753.210
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	5.135	2.922	5.755	3760.785	3755.030
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.822	3762.607	3760.785
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3772.107	3762.607
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.625	3773.732	3772.107
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.19	3774.922	3773.732
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.7	3776.622	3774.922
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.27	3785.892	3776.622
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.725	3787.617	3785.892
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD	4/00	2.922	0.23	3787.847	3787.617
	31	7" 29 PPF CASING HYDRAULIC SET RERMANENT PRODUCTION PACKER TO BE RUN IN 7.00" 29 PPF CASING	5//910	2.922	4.44	3792.287	3787.847
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.250	2.922	1.822	3794.109	3792.287
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	3803.609	3794.109
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.825	3805.434	3803.609
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.562" TYPE XN (NO-GO ID: 2.442")	4.5	2.562	0.37	3805.804	3805.434
	38	RE-ENTERYY MULE SHOE	4.25	2.922	0.37	3806.174	3805.804
		END OF COMPLETION STRING					3806.174

Tubing Retrievable Saftey Valve	66.060	64.280
Traveling Joint	3760.785	3755.030
CIV 1	3728.040	3727.310
CIV 2	3741.910	3741.180
SSD	3774.922	3773.732
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER	3792.287	3787.847
NO-GO Landing Nipple	3805.804	3805.434
Entry Guide	3806.174	3805.804

Figure 7: Proposed Completion Schematic

9. BOP, Wellhead & X-Mass Tree

9.1 Existing Wellhead & X-Mass Tree

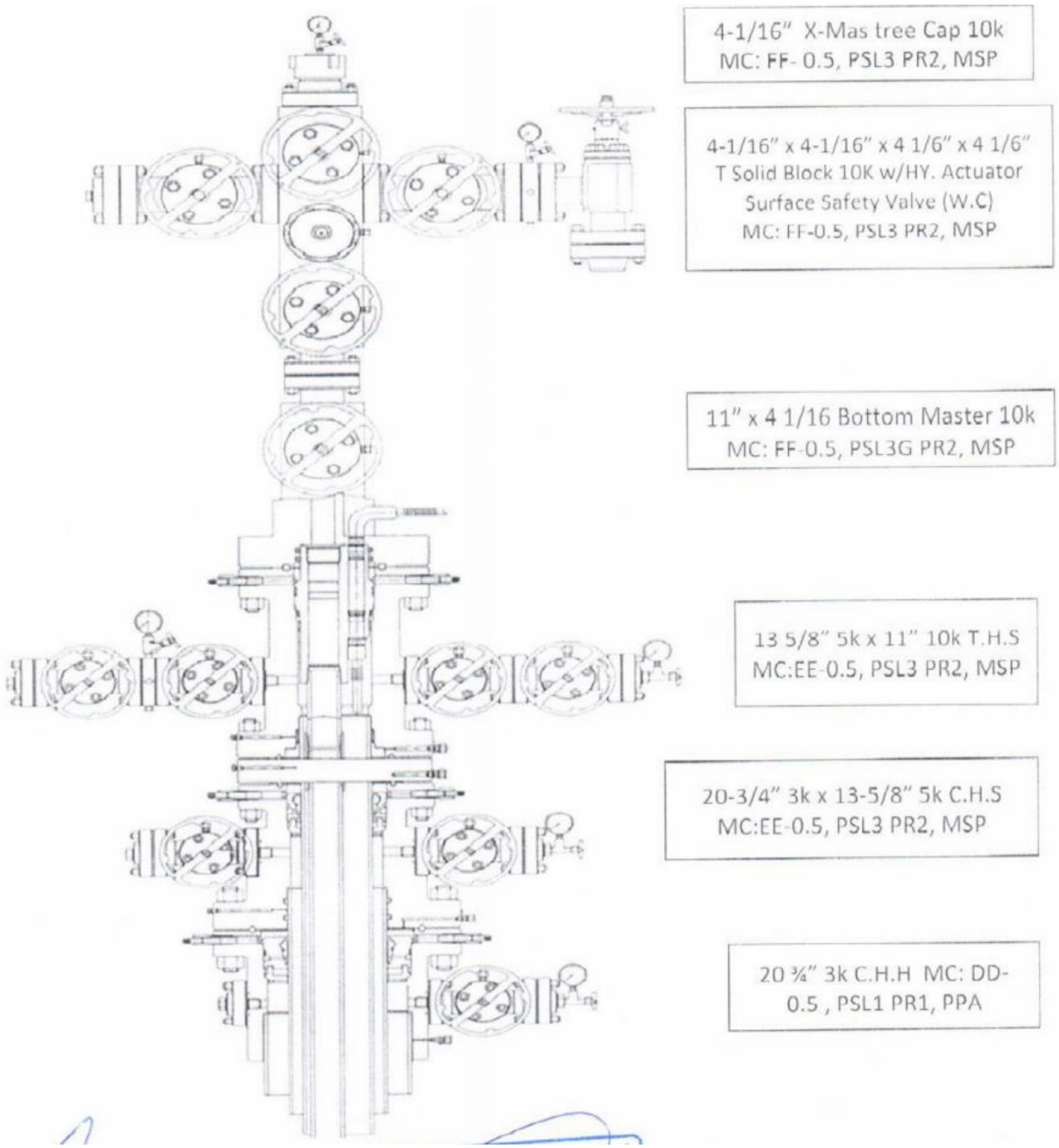


Figure 8: Existing Wellhead

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Sarvak	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Sarvak & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Sarvak & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Table 10: AZNS Wellhead Information

- **Wellhead Specification**

NOTE: Wellhead and X-Mass Tree already installed by client.

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm**

	Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Table 11: AZNS 122 Wellhead Specification

9.2 BOP

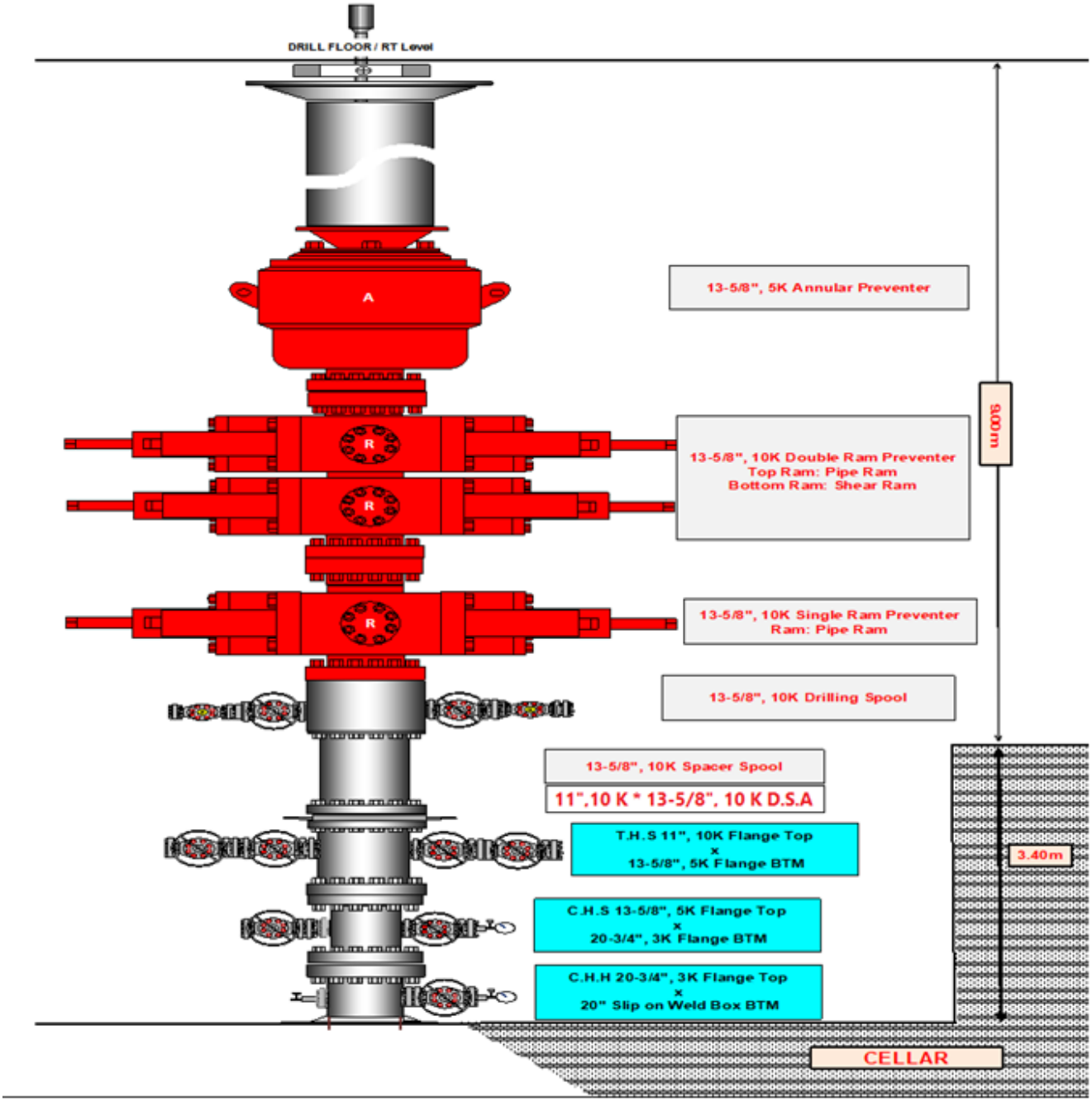


Figure 9: BOP Sketch

10.

10. Waste Management

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

11. Health, Safety and Environments

12.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA and contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

12.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

12.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

12.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System and well control equipment shall be recorded in the IADC daily reports and shall be signed by both the Drilling Contractor's Tool Pusher and Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

12.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.
While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

12.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.
Extreme care shall be taken to monitor mud volume, drilling breaks and gas cut mud.
The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.
The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.
The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.
The Driller is responsible for carefully measuring and recording the slow pump pressure.
Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure and record the slow pump pressure at three different flow rates with all pumps at the following times:

Once per tour.

Whenever changes occur in the mud density/rheology.

Whenever changes occur in the dimension and characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

12.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop and a static monitoring of the well shall be carried out. Space-out for all Pipe Rams and Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer and rams.

12.6 Well control during drilling and completion operation

The drilling design shall be strictly implemented by the drilling contractor and when it is discovered there is incompliance of downhole situation with design and the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, and safe operation can be affected, after the discussion between drilling rig manager and the drilling supervisor, practical technical measures can be taken first and then report to the top management. The Company shall approve and notice in time as per the drilling program amendment procedure.

The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program and curve of actual mud weight must be drawn during drilling operation. For exploratory well and key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone and shale shall be drawn by the comprehensive mud logging crew.

When entering oil and gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing and mud loss.

SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate and circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate and circulating pressure shall be recorded in Daily Drilling Report and Kill Sheet.

Once for every 12 hours in general;

After bit change or BHA change; 3) Mud weight change exceeds $^{3.75}$pcf.

Oil and gas invasion and overflow shall be checked with wiper trip and shall be conducted under the following situation:

Each time before POOH after drilling oil and gas formations.

Severe invasion of oil and gas occurs during drilling, but overflow does not occur before POOH.

POOH after killing well.

POOH after lowering mud weight in wellbore.

POOH after curing mud loss during drilling oil and gas formation.

POOH after drilling oil and gas formation and a prolonged period of time is needed for other operations (E logging, running casing, running tubing and DST etc.).

Regarding the shut-in methods selection

Soft shut-in method is recommended for gas well;

Hard shut-in method is recommended for oil well;

Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;

Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.

Flow check shall be conducted regularly via trip tank as follows

Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)

POOH to last casing shoe

Before POOH BHA through BOP stack

Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.

During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.

When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.

During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.

During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. if equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.

Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud. Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.

The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.

During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if

special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m^3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

12.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.

Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

12.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.

Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.

All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)

All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.

PSSR (Pre-Startup Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

12.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.

Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.

Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.

Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)

Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.

Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.

The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.

Local community are recommended.

Hazardous Wastes

Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site

Mud purifying circulation system.

Cooling water spraying and circulating system for drilling pump.

Special recovery tank for deposed oil.

Reserve house for valuable mud chemicals.

Automatic mud weighting supply system.

BOP stack is equipped according to drilling design.

Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location

Sewage pit Area of the sewage pit: 1500 m2.

Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.

Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.

Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.

Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.

Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.

After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.

Install fuel supply lines and keep them in good condition.

All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.

The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:

Paved, fenced, marked and illuminated

Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).

In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).

Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).

Capable enough to provide storage for the entire range of hazardous wastes generated during the work.

The area shall be provided with Spill Containment and means of preventing run-off to and from the area.

All waste containers to be stored shall be appropriately labelled.

Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

12. H_2S Monitoring & Prevention Measure

13.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

13.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

13. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

14. Requirements

Item	Description	QTY.	Unit	Supplier
1	8-1/2" Milled Tooth Bit	1	Ea.	MSA
2	5-7/8" or 6" Mill	5	Ea.	MSA
3	2-1/4"*4" Taper Tap	2	Ea.	MSA
4	4 1/2" * 3 1/2" Completion String 10K	1	set	MSA
5	9-5/8" & 7" Scraper	1+1	ea.	OEOC
6	5" DP,HWDP,8-1/2" DC,	as required		OEOC
7	Well Head Services	as required		Service Co.
8	Surface Well Test Services	as required		Service Co.
9	Completion Services	as required		Service Co.
10	Slick line Services	as required		Service Co.
11	Casing Running Services	as required		Service Co.

12	Wireline Logging	as required		Service Co.
13	Dry Test, DST, RTTS	as required		Service Co.
14	Cementing Services	as required		Service Co.
15	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
16	H2S Services	as required		Service Co.
17	Mud Logging Services	as required		Service Co.
18	Rig Services	as required		Service Co.
19	Drilling & Completion Fluid Services	as required		Service Co.
20				

Note: Quantity & Spec to be conformed with contract's SOW.
Note: In case of any shortage or discrepancy, the contract's SOW will be reference.

Table 12: List of Requirements

15. Revision Corrections

Drilling Engineer							
1	**Well Information:** • مدت زمان مشخص شده برای contingency plan اصلاح شود و به ۵۸,۹ تغییر کند.	7	CM	اصلاح گردید			
2	**Casing & Liner Depth:** • Grade آستری ۷ اینچ مجددا بررسی گردد و اصلاح شود.	50	CM	اصلاح گردید			
3	**Time break down:** • بعد از جریان دهی چاه عملیات ذکر شده بعدی نا خوانا می باشد، اصلاح شود. • زمان ها در contingency plan اصلاح شوند • Contingency plan #5 به صورت زیر اصلاح شود: **Contingency plan #5: as per client request perform dry test and related to result run and set SN. Packer.**	57	CM	اصلاح گردید			
4	**Sequence of Operation:** • Contingency plan #5 به صورت زیر اصلاح شود: **Contingency plan #5: as per client request perform dry test and related to result run and set SN. Packer.**	59-61	CM	اصلاح گردید			
Fluid							

	لیست تجهیزات و آزمایشگاه پسماند در برنامه قید گردد.		CM	در برنامه مدیریت پسماند بصورت جداگانه ارسال می گردد			
Geology							
	صفحه ۱۴ برنامه چاه، اطلاعات فشار Current با شرح کار همخوانی ندارد.Reservoir Pressure	14	CM	اصلاح گردید			